US-China Decoupling and Board Composition: Evidence from Chinese Listed Firms

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Abstract

As the United States and China decoupled, U.S.-connected directors left the boards of Chinese listed firms under the personal legal exposure that U.S. sanctions created and the Persons Rule sharpened. Treating the sanctions episode as quasi-exogenous, we identify the board response from a director-level within-board exit hazard that compares each U.S.-connected director only with the non-U.S. colleagues on the same board in the same year, using an auditable three-prong U.S.-nexus definition built from structured records; a firm-level replication in Hong Kong traces the same mechanism. After the 2021 sanctions the within-board exit gap widens to roughly 4.6 percentage points by 2022, and exposed mainland firms shed about two percentage points of U.S.-nexus director share, deepening from 1.3 to 2.9 percentage points under the conduct-based Persons Rule. The exit concentrates among independent directors, operates through non-renewal at term end, and is confined to U.S.-connected directors, while substitution is market-specific: mainland boards recruit domestically credentialed professionals, and Hong Kong boards recruit non-U.S. internationals. The pattern is consistent with an individual-liability compliance chill. Governance metrics show no near-term deterioration, while a modest Tobin's Q decline at exposed private firms suggests that the departing directors' international human capital was not fully replaced.

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1 Introduction

In recent years, prominent Chinese listed firms have watched their U.S.-connected directors leave their boards as Washington and Beijing have pulled their economies apart. In March 2023, for instance, two naturalized U.S. citizens, Du Zhiyou and Huang Qing, resigned from the board of Advanced Micro-Fabrication Equipment (AMEC), a Shanghai-listed maker of chip-manufacturing equipment, citing personal reasons only months after Washington had barred "U.S. persons" from supporting advanced chipmaking in China.[1] The firm's own founder and chairman, a Silicon Valley veteran of Intel and Applied Materials, would himself renounce his U.S. citizenship and reclaim Chinese nationality.[2] A co-founder of Arm (a British chip designer), who resigned from the board of Semiconductor Manufacturing International Corporation (SMIC) after nine years, observed that "the international divide has further widened."[3] Over this period, the United States and China have undergone a marked process of economic and financial decoupling, with a growing literature documenting its consequences for trade, innovation, and firm value (Egger and Zhu, 2020; He et al., 2022).

A fundamental question nonetheless remains unanswered: through what internal governance mechanisms do geopolitical shocks reshape firms? This Article addresses that question by examining how U.S.–China decoupling has reshaped board composition in Chinese listed firms, and by asking who replaces the directors who depart. We document board reconstitution as a previously underexamined margin along which geopolitical fragmentation reaches the firm, a margin that operates through the market for director human capital and that is distinct from the trade and financial channels emphasized in prior work.

Our analysis proceeds in two steps. First, we ask whether U.S. sanctions reshape board composition, that is, whether they reduce the share of directors with U.S. backgrounds on the boards of affected Chinese firms, and, when a seat is vacated, whether it is filled by a domestic professional, by a state-affiliated figure, or left empty. We then ask whether the same shock reshaped boards in an independent cross-market setting, namely Hong Kong-listed Chinese firms, where the foreign investor base is directly measurable and its response can be observed.

Answering these questions requires a variable that standard corporate-governance data do not report—the nationality and U.S.-nexus of individual directors—which must be constructed from director-level records. We anchor an auditable three-prong U.S.-nexus definition (i.e., U.S. nationality or permanent residency, substantial U.S. work experience, or U.S. professional licensure) in structured records from the China Stock Market and Accounting Research (CSMAR) database,

[1] "Tech war: US directors resign from board of Shanghai-listed chip equipment firm amid Washington restrictions on citizens," *South China Morning Post*, 15 March 2023, https://www.scmp.com/tech/tech-war/article/3213626/tech-war-us-directors-resign-board-shanghai-listed-chip-equipment-firm-amid-washington-restrictions.

[2] "Tech war: founder of Chinese semiconductor equipment maker AMEC renounces US citizenship," *South China Morning Post*, 18 April 2025, https://www.scmp.com/tech/tech-war/article/3307075/tech-war-founder-chinese-semiconductor-equipment-maker-amec-renounces-us-citizenship.

[3] "Ex-Arm boss exits Chinese chipmaker amid US-China tensions," *Al Jazeera*, 11 August 2022, https://www.aljazeera.com/economy/2022/8/11/ex-arm-boss-exits-chinese-chipmaker-amid-us-china-tensions; "Tudor Brown resigns from SMIC," *Electronics Weekly*, 11 August 2022, https://www.electronicsweekly.com/news/business/tudor-brown-resigns-from-smic-2022-08/.

and for directors whose connections are not verifiable from structured fields we supplement it with web-scraped biographical text classified by a large-language model, used only in the robustness treatment definitions and cross-checked against an independent nationality classifier; the Hong Kong arm instead rests on surname- and nationality-based classifiers suited to that market's thinner structured coverage. We treat this measurement as the methodological foundation for the analysis, with its full construction and validation reported in Appendix A.1.

Our central finding is that U.S.–China decoupling drove U.S.-connected directors from the boards of exposed Chinese firms. The identifying evidence is a within-board comparison: on the same board in the same year, U.S.-connected directors depart at sharply higher rates than their non-U.S. colleagues, a gap that widens to roughly 4.6 percentage points in 2022 against a mean annual exit rate near 14.5 percent. In the aggregate this exit reduces the proportion of directors with U.S. backgrounds by approximately two percentage points, roughly one-third of the pre-shock treated-firm mean, and the loss roughly doubles between the investment-sanctions era and the Persons-Rule era (Section 4.2). The vacated seats are refilled one-for-one, overwhelmingly by domestically credentialed professionals, and departure occurs through passive non-renewal at term end, with forced mid-term resignation playing little role. We interpret the exit as causal—subject to the threats to identification we take up in Section 7—and the replacement pattern as descriptive, a distinction Section 3 formalizes.

The Article makes two contributions, unified by a single frame: one mechanism, operating at two intensities of legal exposure, across two markets. The mechanism is an individual-liability compliance chill—the heightened personal legal and reputational risk under the U.S. Persons Rule that raised the cost of board service for U.S.-connected directors even where no firm-specific ban applied—and its intensity runs from intent-to-treat exposure, under which the response is selective, to explicit sanction, under which it becomes indiscriminate. The first contribution isolates the mechanism at its selective pole. The exit appears across the full set of firms with a U.S.-connected director, including firms under no direct legal prohibition, the breadth that marks it as compliance chill rather than compliance with a specific ban. A within-firm placebo separates this U.S.-specific channel from a concurrent market-wide retreat of foreign directors from Chinese boards, because at exposed firms the elevated exit is confined to U.S.-connected directors, while non-U.S. foreign directors are unaffected on the mainland and actively recruited in Hong Kong, and firms with non-U.S. foreign exposure but no U.S. connection lose those directors regardless of sanctions. To our knowledge this is the first paper to isolate the sanctions-specific component of board de-internationalization from the broader Sino-Western governance fragmentation trend. A decoupling literature has documented broad stock-market reactions and spillovers from U.S.–China tensions (Egger and Zhu, 2020; He et al., 2022; Oh and Kim, 2024) and has shown that sanctions and export controls reshape cross-border investment and strategic adjustment (Biglaiser and Lektzian, 2011; Hu et al., 2024; Liu and Li, 2023), yet it has largely left unopened the internal channel through which those effects reach the firm; a complementary strand studies a different such channel, U.S. asset managers' proxy voting on charter amendments that formalize Chinese Communist Party committees (Lin, 2026; Lin and Milhaupt, 2021), whereas the channel we trace operates through the market for director human capital. Because the departing directors were disproportionately independent monitors, the loss carries governance stakes that Section 6 takes up directly.

The second contribution is cross-market replication that traces the mechanism across its intensity range. Extending the analysis to Hong Kong-listed Chinese firms, where the foreign investor base is directly measurable, reproduces the director-exit result in a distinct regulatory and ownership environment and shows that substitution is market-specific: mainland boards draw on a domestic talent pool, whereas Hong Kong boards draw on a deep non-U.S. international pool. At the indiscriminate pole, the explicitly sanctioned Hong Kong firms, eleven in all, show the exit widening to foreign directors of all origins together with measurable capital reallocation, while at intent-to-treat firms international and mainland ownership show no detectable movement beyond a Stock Connect decline; we report that capital-base evidence in Appendix A.3 and read it cautiously, since the mainland and intent-to-treat ownership nulls partly reflect low baseline foreign exposure and therefore bear weakly on investor behavior.

These questions also carry policy implications. By documenting the exit of U.S.-connected directors and the domestic professionals who replaced them, this Article helps policymakers assess how geopolitical fragmentation reshapes corporate boards and how deep the domestic director-labor market is that absorbs such shocks. The pattern also points to an unintended consequence of extraterritorial sanctions. Because the directors who departed were disproportionately independent directors with international credentials—the type that a substantial literature associates with stronger monitoring and with constraining controlling-shareholder expropriation in Chinese firms (Zhu et al., 2016; Gong et al., 2021; Jiang et al., 2010)—a measure aimed at restricting U.S. capital flows may have incidentally removed a potential source of minority-shareholder protection from the affected boards. Whether that removal imposed an actual cost on domestic investors remains an open question: our consequence analysis in Section 6 detects no measurable near-term deterioration in crash risk, in the market's reaction to the departures, or in executive pay-performance sensitivity and payout policy, a bounded null we read alongside the one-for-one domestic substitution. The one margin on which a cost registers is valuation: Tobin's Q declines at exposed privately owned firms, most clearly after the departure of postgraduate-credentialed directors, while the state sector shows no comparable movement, so the measurable cost of the sanctions fell on private firms rather than on the state-linked entities the measures targeted. Longer-horizon evidence would be needed to settle the point within the broader policy conflict between the sanctioning and sanctioned governments.

The remainder of the Article proceeds as follows. Section 2 describes the U.S. sanctions framework and the individual-liability channel and reviews the related literature. Section 3 sets out the data, the measurement of U.S.-nexus board connections, and the empirical strategy, whose causal weight rests on within-firm and within-board comparisons. Section 4 reports the exit of U.S.-nexus directors, the board reconstitution that follows, and the placebo decomposition that isolates the U.S.-specific channel. Section 5 presents the cross-market replication in Hong Kong. Section 6 examines whether the loss of these directors matters for firm governance and firm value, finding preserved monitoring outcomes alongside a suggestive valuation decline at privately owned firms. Section 7 discusses limitations and extensions, and Section 8 concludes.

2 Institutional Background and Related Literature

2.1 Sanctions and Individual Liability

The decoupling pressure that concerns this Article operated through two families of federal measures: investment sanctions administered by the Treasury, which restrict what U.S. persons may own, and export controls administered by the Commerce Department, which restrict what U.S. persons may do. The two arrived in sequence across the Trump and Biden administrations, and each contributes a distinct layer of the individual-liability environment this section describes.

The investment track began in November 2020, when President Trump signed Executive Order 13959, prohibiting any "U.S. person" from purchasing the publicly traded securities of companies that the Department of Defense had designated "Communist Chinese military companies," a roster of prominent firms—China Mobile, CNOOC, and SMIC among them—that the Pentagon had begun publishing that summer.[4] The Biden administration preserved the policy while rebuilding its legal architecture: Executive Order 14032, signed on June 3, 2021, moved the designation authority to the Treasury, whose Office of Foreign Assets Control (OFAC) maintains the operative roster as the Non-SDN Chinese Military-Industrial Complex Companies (NS-CMIC) List, reset the designation criteria to reach firms operating in China's defense-and-related-materiel or surveillance-technology sectors, and prohibited U.S. persons from purchasing or selling the publicly traded securities of the 59 companies in its initial annex, with the prohibitions taking effect on August 2, 2021.[5]

These prohibitions reach directors because of whom they bind. The sanctions regime construes a "U.S. person" broadly, reaching any U.S. citizen, lawful permanent resident, entity organized under U.S. law (including its foreign branches), and any individual physically present in the United States, irrespective of where the person resides or where the conduct occurs. A director who falls within this definition is bound by the prohibitions personally, independent of the nationality of the firm on whose board the director sits.

For a U.S.-person director of a designated firm, individual exposure arises even though board service as such is not enumerated among the prohibited acts. The regulations separately prohibit any transaction that evades or avoids, has the purpose of evading or avoiding, or causes a violation of the prohibitions, together with conspiracies and attempts to do so, so that a director who

[4] Exec. Order No. 13,959, 85 Fed. Reg. 73,185 (Nov. 17, 2020). The Department of Defense issued the designations under § 1237 of the National Defense Authorization Act for Fiscal Year 1999 in tranches of June 24, August 28, and December 3, 2020, and January 14, 2021. The purchase prohibition took effect on January 11, 2021, with divestment authorized through November 11, 2021; Executive Order 13,974 of January 13, 2021, 86 Fed. Reg. 4,875, adjusted the divestment mechanics. An earlier order, Executive Order 13,873 of May 15, 2019, 84 Fed. Reg. 22,689, had declared a national emergency over the information-and-communications-technology supply chain and authorized the Commerce Department to prohibit such transactions with "foreign adversaries"; it named no listed firm and imposed no securities restriction, and we treat it as background.

[5] Exec. Order No. 14,032, 86 Fed. Reg. 30,145 (June 7, 2021). OFAC's list entries carry a uniform effective date of August 2, 2021, with divestment authorized through June 3, 2022. Subsequent additions, including DJI and SenseTime on December 16, 2021, brought the roster to roughly seventy firms. OFAC codified the prohibitions as the Chinese Military-Industrial Complex Sanctions Regulations, 31 C.F.R. part 586, in February 2022.

approves or authorizes a covered securities transaction by the firm, such as an issuance, a buy-back, or treasury dealing in covered instruments, may thereby cause or facilitate a violation even without trading on personal account. Equity-linked director compensation, including shares, options, or restricted units in a covered firm, can itself constitute the acquisition of a prohibited security. And because civil liability under the International Emergency Economic Powers Act attaches on a strict-liability basis, without proof of willfulness, a director's good-faith uncertainty about whether a particular board decision touches a covered security does not avert exposure.[6] The reach of this track is nonetheless confined to the designated firms; a U.S.-person director of a Chinese listed company outside the NS-CMIC List faced no prohibition from it.

The conduct track arrived in October 2022, bears on individuals more directly than the investment sanctions do, and is the measure this Article refers to throughout as the U.S. Persons Rule. On October 7, 2022, the Commerce Department's Bureau of Industry and Security (BIS) issued sweeping export controls on advanced computing and semiconductor manufacturing, among whose provisions, effective October 12, 2022, was a requirement that any U.S. person obtain a license, reviewed under a presumption of denial, before "supporting" the development or production of advanced-node integrated circuits at any facility in China.[7] Three features gave the rule its force for board service. It governs conduct, so the license requirement attaches to a person's work—servicing equipment, transferring technology, facilitating production—rather than to any securities transaction. It applies by sector, reaching any Chinese firm engaged in advanced-node semiconductor production whether or not the firm appears on a sanctions list. And its penalties are criminal as well as civil. A U.S.-citizen or permanent-resident director or officer of such a firm accordingly could no longer separate continued service from the firm's now-restricted core activity, and the consequences were immediate: within weeks, American executives across Chinese chip firms suspended work or resigned, and the AMEC resignations with which this Article opens followed within months.[8]

Each track therefore binds its own subset of the directors we observe: the securities prohibitions reached U.S. persons at the designated firms from August 2021, principally through equity-linked compensation and board approval of covered transactions, while the U.S. Persons Rule reached

[6] 31 C.F.R. §§ 586.201, 586.701; 50 U.S.C. § 1705. Civil penalties reach the greater of a statutory maximum adjusted annually for inflation (approximately $368,000 per violation as of 2024) or twice the value of the offending transaction; willful violations carry criminal penalties of up to $1,000,000 and twenty years' imprisonment. Holding a covered security acquired before designation is not itself prohibited, and time-limited general licenses authorized wind-down and divestment, which concentrated the practical pressure to exit covered positions around the 2021 effective dates.

[7] 87 Fed. Reg. 62,186 (Oct. 13, 2022), amending 15 C.F.R. § 744.6. "Support" comprises shipping, transmitting, or transferring covered items, servicing them, or facilitating such activity, where the person knows the items will be used to develop or produce integrated circuits at a Chinese fabrication facility meeting specified technology thresholds: logic chips at the 16/14-nanometer node or below, NAND flash memory at 128 layers or more, or DRAM at an 18-nanometer half-pitch or below. The controls rest on the Export Control Reform Act of 2018, 50 U.S.C. § 4801 et seq. Under 50 U.S.C. § 4819, willful violations carry criminal penalties of up to $1,000,000 and twenty years' imprisonment, and civil penalties, which require no showing of willfulness, reach the greater of $300,000 or twice the transaction value per violation. Rules of October 17, 2023, 88 Fed. Reg. 73,424 and 88 Fed. Reg. 73,458, tightened the same controls.

[8] "American Executives in Limbo at Chinese Chip Companies After U.S. Ban," *Wall Street Journal*, 16 October 2022; contemporaneous reporting in the *South China Morning Post* described equipment makers, including NAURA, withdrawing U.S.-passport staff from restricted development work in the days after the rule issued.

U.S. persons connected to advanced semiconductor production at any Chinese firm from October 2022, through their work itself. The contemporaneous tightening of the BIS Entity List, which restricted designated firms' access to U.S. technology, bound U.S. exporters rather than directors personally and forms part of the surrounding pressure.[9] Neither instrument reached every firm at which we observe U.S.-connected directors, and in that gap lies the exposure this Article exploits: the combination of expansive personal jurisdiction, civil liability without willfulness, and criminal tail risk raised the expected private cost of board service at politically exposed Chinese firms generally, well beyond the instruments' literal commands. This role-, compensation-, and conduct-contingent exposure is the source of the individual-level legal and reputational risk that we later term compliance chill. The exposure is most acute for independent directors, who typically receive the fee- or equity-based compensation that falls within the securities regulations' reach and who, because they lack an executive or controlling stake, bear lower switching costs than insider directors once the legal cost of continued service rises. Beyond these direct channels, the increasingly hostile geopolitical environment may also have encouraged foreign directors, as well as internationally connected directors based in China, to resign from Chinese boards even absent any applicable prohibition, since the reputational and regulatory costs of such associations grew harder to justify. The mechanism described above connects to two strands of prior work, one on the geopolitics of board composition and one on foreign-director human capital, which we review in turn.

2.2 U.S.–China Decoupling and Board Composition: Exit, Replacement, and Domestic Professionalization

Recent evidence suggests that director retention is sensitive to the geopolitical environment, because board service is embedded in cross-border legal, reputational, and identity contexts. As Tian et al. (2025) document, foreign directors are more likely to exit Chinese boards when bilateral political relations deteriorate. Complementing this identity-based mechanism, Liu et al. (2025) show that sanctions can generate "shock anticipation" in director labor markets, so that foreign directors may resign when their own firms are directly targeted and when sanctions imposed on peer firms raise perceived legal and reputational risk. This anticipatory resignation logic is consistent with the broader finding that outside directors tend to quit preemptively when they foresee reputational or legal exposure ahead (Fahlenbrach et al., 2010).

A board human capital perspective, however, requires that we move beyond exit alone (Adams et al., 2010). Because board seats are typically refilled, the identity and expertise of replacement directors can meaningfully change a board's human capital and governance capacity. Indeed, a growing literature on China suggests that specific kinds of professional directors matter for governance. Zhu et al. (2016) show that more empowered independent directors engage in stronger monitoring and are associated with higher firm value, while Chen et al. (2019) show that forced resignations of academic independent directors in China reduce firm value, which

[9] Huawei was added to the Entity List on May 16, 2019, 84 Fed. Reg. 22,961, and SMIC on December 18, 2020, 85 Fed. Reg. 83,416, the latter with a presumption of denial for items enabling production at advanced technology nodes.

suggests that academically trained directors contribute economically meaningful expertise. As He and Liu (2016) document, Chinese listed firms appoint independent directors with professional legal backgrounds for specific consulting and governance purposes, thereby treating independent directors as heterogeneous sources of expertise. More broadly, Firth et al. (2016) show that regulatory sanctions imposed on independent directors in China have persistent consequences for their future board opportunities, which reinforces the plausibility of systematic director-labor-market adjustment after external shocks.

Regulation can also change board composition directly by determining which types of directors may serve. Fan (2021) studies China's regulation banning politicians from corporate boards and shows that this external regulatory shock led to substantial board reconfiguration and affected firm performance. Berkowitz et al. (2022) extend this logic, showing that firms that lost politically connected directors initially suffered a valuation decline but subsequently became more productive, transparent, and innovative, a pattern of short-run disruption followed by longer-run governance improvement. Both papers demonstrate that external political and regulatory shocks can reshape the mix of directors on boards, and that the consequences depend critically on who replaces those who depart. Building on these insights, this Article examines how U.S.–China decoupling pressures drive the exit of U.S.-nexus directors from Chinese listed firms, who replaces them, and what that substitution implies for overall board human capital. In particular, we examine whether firms respond by substituting toward domestically credentialed professional directors when U.S.-connected board human capital becomes more costly to maintain.

2.3 Board Composition and Foreign-Director Human Capital

A large literature studies how board composition affects firm governance and firm value (Adams et al., 2010). The board internationalization literature argues that foreign directors affect firm outcomes through both monitoring and advising channels, including the transfer of expertise and the reduction of information frictions (Masulis et al., 2012; Oxelheim and Randøy, 2003; Oxelheim et al., 2013). Related work suggests that foreign independent directors may be especially valuable where domestic legal institutions are weaker, or where the directors themselves come from stronger legal environments (Miletkov et al., 2017). Evidence from China similarly suggests that directors with overseas experience improve firm performance and the information environment (Giannetti et al., 2015), that foreign experience is associated with lower stock-price crash risk (Cao et al., 2019), and that it is associated with different payout policies (Tao et al., 2022). A related external-certification view argues that foreign-connected directors can enhance credibility with global investors and transaction partners, especially where liability-of-origin concerns are salient (Yu et al., 2024). More broadly, the cross-listing literature establishes the general principle that firms with tighter links to developed-market capital and governance standards command a valuation premium, because those links signal commitment to outside investors (Doidge et al., 2004). Because the departure of U.S.-connected directors in our setting is driven by an external political shock, and is therefore separated from endogenous board choices, we can study how boards recompose when this internationally connected human capital becomes costly to retain, thereby distinguishing the exit from firms' ordinary board-selection decisions.

The Chinese governance literature also suggests that director quality matters for monitoring and value. As Liu et al. (2015) show, board independence has a positive overall effect on firm performance in China; Jiang et al. (2016) document that independent directors' reputation concerns shape their voting behavior; and Gong et al. (2021) find that stronger board independence can reduce tunneling by controlling shareholders. The tunneling channel is especially consequential in the Chinese context, where controlling shareholders have historically used intercorporate loans and related-party transactions to siphon resources from listed companies at the expense of minority shareholders (Jiang et al., 2010).

Two features of this prior work motivate our design. Most corporate-governance research treats board structure as the outcome of firm-level economic choices, such as the need for expertise, monitoring, or external resources (Adams et al., 2010), whereas the sanctions episode we study is an external political shock that alters board composition for reasons unrelated to firms' underlying economic fundamentals. Because boards are chosen by firms and directors self-select onto them, board-composition studies routinely face an endogeneity problem that an externally imposed, sanctions-driven departure helps to overcome, and that opening is what this Article exploits in tracing how a geopolitical shock reshapes the internal organization of Chinese listed firms.

Figure 1 consolidates this Article's identification logic and the compliance-chill mechanism in a single diagram. It traces the quasi-exogenous sanctions shock, the individual-liability channel through which it operates, the director exit we interpret as causal, the descriptive board recomposition that follows, and the selective-versus-indiscriminate threshold around which the cross-market evidence is organized.

3 Data and Empirical Strategy

3.1 Measuring U.S.-Nexus Board Connections

The director-level attribute on which the analysis turns—whether a director holds a meaningful connection to the United States—is not reported in standard corporate-governance data, so we construct it here. Our primary measure classifies a director as U.S.-nexus if any of three conditions holds in CSMAR structured records: U.S. nationality or permanent residency, at least five years of U.S. work experience, or a U.S. professional license, applied to the resume, education, professional-title, nationality, and licensure fields of the CSMAR Corporate Governance module. Because this structured measure is fully auditable and relies on no model-based inference, we use it as the regressor underlying all headline estimates. For directors whose connection cannot be verified from structured fields, we supplement it with a classification layer over web-scraped biographies that defines the expanded treatment definitions used in robustness checks; in the Hong Kong market, where structured nationality coverage is thinner, we infer director origin from a multi-dialect Chinese surname classifier validated against an independent nationality classifier. Appendix A.1 sets out this measurement hierarchy and its validation, Appendix A.2 details the nationality-classification procedure itself, and the headline exit estimate is essentially unchanged

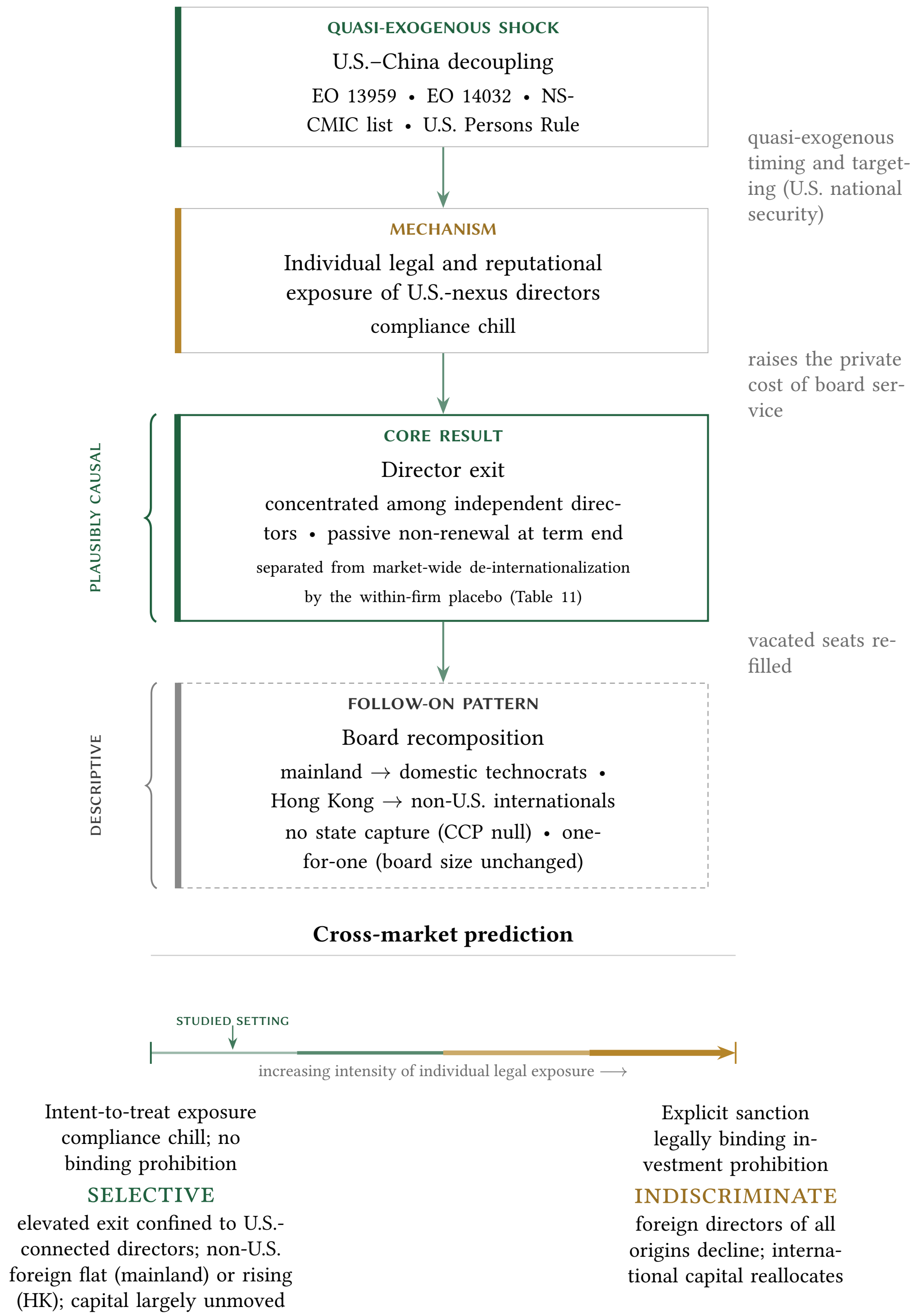

Figure 1: Identification and mechanism. The vertical chain traces the causal architecture: a quasi-exogenous U.S. sanctions shock raises the individual legal and reputational exposure of U.S.-nexus directors (compliance chill), which we interpret as causing the director exit, around which boards descriptively recompose. The within-firm placebo decomposition (Table 11)—under which the elevated exit is confined to U.S.-connected directors while non-U.S. foreign directors at the same firms show no comparable departure—separates the U.S.-specific channel from a concurrent market-wide de-internationalization, while the director-level within-board exit hazard, which compares U.S.-connected directors with their boardmates, is the design least vulnerable to firm-level mean reversion. The lower panel states the cross-market prediction: the response is selective under intent-to-treat exposure and becomes indiscriminate once an explicit sanction renders covered securities transactions legally prohibited.

once classified directors are added (Table A6).

3.2 Data and Sources

We procure our director and resume data from two CSMAR modules: the Corporate Governance module (annual snapshots 2010–2024) and the director changes file (event-level). Because the primary Corporate Governance export reaches the vendor's row cap before covering the highest-numbered stock codes, we combine it with a supplementary pull covering the remaining listed firms, and we measure the nationality and background fields identically across the two exports, so that a director's U.S.-nexus classification does not depend on which export supplied the record. The combined roster contains 87,208 distinct directors across 2,920 firms and 374,614 director-firm-years over 2018–2024. The annual file supplies resume text, education background, professional titles, and overseas work and education indicators, while the event file records appointment and departure dates alongside resignation-reason text; from these fields we construct the U.S.-nexus measure described above.[10]

The remaining firm-level variables come from three further CSMAR modules. Following the standard governance-panel practice, we draw our financial controls—log assets, leverage (i.e., total liabilities over total assets), and ROA (i.e., net income over total assets)—from the Financial Statement modules, where they serve as time-varying covariates winsorized at the 1st and 99th percentiles. The Ownership and Control module supplies a time-varying state-owned/private (SOE/POE) classification together with top-10 shareholder concentration, while the board and compensation variables—board size (i.e., the board-of-directors count), the independent-director ratio, top-three executive pay, and aggregate institutional ownership—come from the Executive Compensation and Governance module, whose coverage exceeds 99% over the sample period.

Our estimation sample comprises 2,911 unique firms (1,040 treated and 1,871 control) observed over 2019–2024, yielding 16,987 firm-year observations. We classify a firm as treated if it had at least one U.S.-nexus director in the 2019–2020 baseline; control firms are matched by industry (China Securities Regulatory Commission, or CSRC, two-digit) and internationalization status, and the post-shock indicator equals one for years from 2021 onward.

3.3 Estimating Equations

To examine how the decoupling shock reshapes board composition, we estimate an exposure-based difference-in-differences (DiD) specification with firm and year fixed effects. For firm j in

[10] The director changes file is itself subject to the vendor's export cap of 1,000,000 rows, which splits it at stock code 600612. As with the roster, we combine the primary export with a supplementary pull covering the remaining listed firms, so all changes-based analyses—the euphemistic-resignation reasons, the director-tenure control, the departure cumulative abnormal returns, and the valuation analyses of Section 6—run on the full two-workbook panel: 87,428 departure events over 2019–2024, of which 60,836 match to the director panel, with the departure-return analysis retaining the 58,800 firm-events that have estimable abnormal returns.

year t,

$$\text{U.S.-nexus share}_{jt} = \beta_1(\text{Treated}_j \times \text{Post}_t) + \gamma X_{jt} + \delta_j + \lambda_t + \epsilon_{jt}, \tag{1}$$

where Treated_j equals 1 for firms with at least one U.S.-nexus director in the 2019–2020 baseline, Post_t equals 1 for years $\geq$ 2021, and X_{jt} collects log assets, leverage, and ROA, while δ_j and λ_t represent firm and year fixed effects, respectively, and standard errors are clustered at the firm level. We expect the coefficient of interest, β_1, to be negative. A dose-response variant replaces Treated_j with the continuous pre-shock intensity $\overline{\text{U.S.-nexus share}}_{j,2019\text{–}2020}$, while an event-study variant replaces $\text{Treated}_j \times \text{Post}_t$ with a full set of year-by-treatment interactions.

The firm-level specification gives the magnitude of the board-composition change, yet, for reasons set out in the identification strategy below, it carries little causal weight, because its treatment is the pre-period level of the outcome. Our identifying design is therefore a director-level within-board exit hazard. For director i on the board of firm j in year t,

$$\text{Exit}_{ijt} = \beta_1(\text{U.S.-nexus}_i \times \text{Post}_t) + \beta_2\,\text{U.S.-nexus}_i + \theta_{jt} + \varepsilon_{ijt}, \tag{2}$$

where Exit_{ijt} indicates that the director leaves the board before year $t+1$, θ_{jt} denotes firm $\times$ year fixed effects, and standard errors are clustered by firm. The firm $\times$ year fixed effects restrict identification to the contrast between U.S.-connected directors and their non-U.S. colleagues on the same board in the same year, so that any firm-level trajectory is differenced out by construction; we expect $\beta_1 > 0$.

We estimate the replacement analysis on matched exit–entry pairs, regressing an indicator for domestic-technocrat replacement on Treated U.S. exit_{ijt} (i.e., a U.S.-nexus director leaving a treated firm post-shock), again with firm and year fixed effects. As Section 4 makes explicit, this analysis is descriptive: it characterizes who fills vacated seats and does not isolate a causal effect of director loss on replacement type. The Hong Kong analyses adopt the same DiD structure, with treatment defined by baseline Western-surname or U.S.-nationality directors, whereas the explicit-sanctions analysis uses a binary indicator for the eleven sanctioned firms.

3.4 Identification Strategy

The exposure difference-in-differences documents the size of the board-composition change. Because its treatment is the pre-period level of the outcome and the sanctions arrived as a single bundled episode, we assign causal weight to the within-firm and within-board-year comparisons—the placebo and the director-level exit hazard—which difference out any firm-level mean reversion by construction. The remainder of this subsection sets out the exposure logic we borrow as motivation, the mean-reversion threat the firm-level design faces, and the two within-unit designs that meet it.

Our identification strategy follows the exposure-based difference-in-differences framework, under which pre-period exposure defines treatment and an external shock supplies the source of quasi-exogenous variation. We define treatment by whether a firm had at least one U.S.-nexus director during the 2019–2020 baseline period, and we aim to identify the effect by exploiting the sanctions shock—driven by U.S. national-security considerations quasi-exogenous to individual firm board composition—as a shift in the costs facing U.S.-connected directors. This structure parallels the "China shock" literature, in which Autor et al. (2013) define treatment intensity by pre-period import exposure and locate identification in the quasi-exogeneity of the trade shock; Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022) provide formal treatments of identification in shift-share designs. Our setting meets the exogenous-shock condition, because the timing, scope, and targeting of U.S. sanctions were determined by Washington's national-security apparatus, independent of the board composition of individual Chinese firms. We borrow this exposure logic as motivation only. Although the license those shift-share designs provide to identify from shock exogeneity in place of parallel trends presumes many independent shocks, our setting rests on a single bundled sanctions episode whose instruments overlap in timing and targeting. With one shock, identification in a difference-in-differences turns on the comparability of treated and control trajectories, the parallel-trends condition that our treatment definition strains; shock exogeneity therefore supplies context for the treatment while leaving the parallel-trends burden in place. The shares themselves are plausibly endogenous to firm characteristics, and the firm and year fixed effects absorb time-invariant share-level confounders. We accordingly borrow the conceptual exposure-DiD logic of the China-shock literature while leaving its large-number-of-shocks asymptotic inference to settings with many independent shocks. This exposure-based design is aimed at the effect of the shock on the measured composition of the board—the outcome to which firms' U.S.-nexus exposure is mechanically linked—and we accordingly limit causal claims to that outcome, treating them as conditional on the identifying assumptions set out below.

The decoupling pressure we study operated as a regime composed of several legal instruments that fired in close succession: Executive Order 13959 (November 2020) and its broadening successor Executive Order 14032 (June 2021), the construction and expansion of the NS-CMIC list, the U.S. Persons Rule, and a contemporaneous tightening of the Bureau of Industry and Security (BIS) Entity List. Because these instruments overlap both in timing and in the firms they reach, we estimate the cumulative response to the bundle; the data cannot separate the marginal effect of Executive Order 14032 from that of the Persons Rule or the Entity List when their effective dates arrive in close succession and the instruments overlap in the firms they reach. We identify the cumulative reduced-form effect of the decoupling regime as a whole, dated to its operative break on August 2, 2021, when the principal investment prohibitions took effect; Figure 2 lays out this legal calendar against the estimation window, and Section 4.2 additionally reports a two-break variant that separates the investment-sanctions era from the Persons-Rule era, asking whether the cumulative response deepened when the conduct track arrived. For the policy question that motivates this Article—how geopolitical fragmentation reshapes corporate boards—the bundle is the relevant treatment, because U.S.-connected directors faced the entire regime. We treat the earlier 2018 trade-war measures, namely the Section 301 tariffs and the first Entity List designations, as a distinct prior shock that the 2019–2024 estimation window deliberately excludes, so that the 2021 break is not conflated with trade-war effects.

Because we define treatment by the level of the outcome variable in the pre-period, treated and control firms are unlikely to share parallel pre-trends in that variable, since treated firms were accumulating U.S.-nexus directors throughout 2015–2019 while control firms had no comparable accumulation. The relevant identifying assumption therefore concerns the quasi-exogeneity of the sanctions shock to pre-period board composition: to establish the causal impact of the decou-

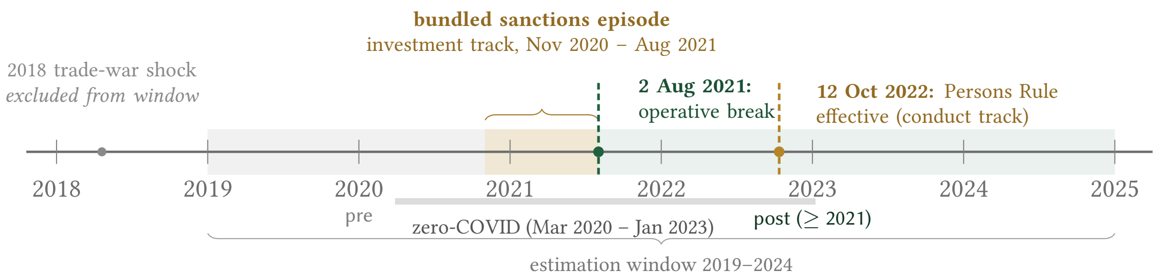

Figure 2: Timeline of the decoupling regime and the estimation window. The treatment is a bundle of U.S. measures whose investment-track instruments fired within roughly twelve months—Executive Order 13959 (November 2020), Executive Order 14032 (June 2021), the NS-CMIC list, and a contemporaneous tightening of the BIS Entity List—with the conduct-track U.S. Persons Rule following in October 2022; their timing and targeting overlap, so the data identify their cumulative effect rather than any single instrument's. We date the pooled treatment to the operative break of 2 August 2021, when the principal investment prohibitions took effect; the second dashed line marks 12 October 2022, when the Persons Rule became effective, the era boundary used by the two-break specification of Table 5. The gray band beneath the axis marks China's zero-COVID border controls, which end before the Persons-Rule-era escalation. The 2019–2024 estimation window excludes the distinct 2018 trade-war shock (the Section 301 tariffs and the first Entity List designations) and affords only a thin pre-period (2019–2020); because identification rests on the within-firm and within-board comparisons, that thin pre-period serves only as a descriptive check.

pling regime, the effect of this shock on board composition must arise solely through its influence on the costs facing U.S.-connected directors and be unrelated to any prior process by which Washington timed or targeted sanctions on the basis of which Chinese firms had Western-connected directors. The principal threat to this design is mean reversion. Because we select firms on a high pre-period U.S.-nexus level, a residual concern is that such firms might mechanically decline toward a common mean regardless of the shock; pre-trend parallelism is unattainable under this treatment definition and therefore cannot serve as the operative assumption. Under a counterfactual scenario where mean reversion drives the post-shock decline, one would expect the U.S.-nexus director share at high-baseline firms to fall back toward the average even absent any sanctions, and one would expect such a firm-level mechanical process to fall on every category of director at those firms. Our central defense against this rival reading is a within-firm placebo that compares U.S.-connected directors with the non-U.S. foreign directors at the same treated firms. Because firm-level mean reversion, mechanical turnover at the end of fixed board terms, and a market-wide de-internationalization trend would each depress all director types at a high-baseline firm, none of them can generate any asymmetry between U.S. and non-U.S. directors within the same firm-years; Section 4 reports this placebo (Table 11).

The resignation analysis embeds this within-firm comparison directly through a triple difference (U.S.-nexus $\times$ Post $\times$ Treated). Several further patterns examined in Section 4 bear on the same comparison: the behavior of the U.S.-nexus share in the single available pre-treatment year, the proportionality of post-shock exit to baseline exposure, and the composition of stated resignation reasons. The dose-response gradient warrants particular care, because a baseline-proportional decline is what mean reversion would also produce; such a gradient corroborates the shock channel without ruling reversion out, the task we reserve for the within-board hazard.[11]

[11]For the same reason—treatment defined by pre-period outcome levels—synthetic control methods (Abadie, 2021) and synthetic difference-in-differences (Arkhangelsky et al., 2021) are inapplicable, as the reweighting procedure

The second, and our primary, defense is the director-level exit-hazard design introduced above, which holds firm and year jointly fixed. The unit of analysis is the director-year and the outcome is whether a director departs the board before the following year, while the specification absorbs firm $\times$ year fixed effects, so that the estimate compares each U.S.-connected director only with the non-U.S. colleagues serving on the same board in the same year. Any firm-level trajectory, including the mean-reverting drift of a high-baseline firm, is differenced out by construction, so that a firm-level pre-trend cannot produce the result. Section 4 reports the estimates from this design (Figure 3 and Table 2). A pre-shock pseudo-treatment built on a 2014–2015 baseline is infeasible in the present panel, which begins in 2018; the within-board design supersedes that check, because it removes by construction the firm-level selection the pseudo-treatment was meant to detect.

A separate question is whether recent critiques of two-way fixed-effects estimators bear on the design. The negative-weighting and contamination problems those critiques identify (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Sun and Abraham, 2021; Borusyak et al., 2022) arise when treatment timing is staggered across units and treatment effects are heterogeneous, so that already-treated units act as controls for later-treated ones. Our setting is largely insulated from this concern, because the sanctions regime is common-timed: the operative break falls in 2021 for every treated firm, leaving no already-treated units to contaminate the comparison, so that the design reduces to a common-timing two-group difference-in-differences outside the staggered case those critiques target. The within-board hazard inherits the same protection, since its firm $\times$ year fixed effects compare directors within a single firm-year and require no cross-cohort comparison. For the continuous dose-response specification, in which treatment intensity varies across firms, Callaway et al. (2024) show that level effects are identified under a generalized parallel-trends condition while the dose gradient itself requires stronger assumptions, so we read the gradient as corroborative evidence that carries no independent identifying force. Should the firm-level Entity List designation dates prove to generate genuinely staggered timing—a refinement we take up in the Discussion—a heterogeneity-robust estimator in the manner of Callaway and Sant'Anna (2021) would be the appropriate response.

We restrict the estimation sample to 2019–2024 to isolate the primary sanctions escalation beginning in 2021 from earlier trade-war measures. The U.S.–China trade conflict escalated in stages: initial Entity List designations beginning in 2018, followed by a relative pause in 2019–2020, and then the major sanctions escalation via Executive Order 14032 in June 2021 and subsequent expansions. Including 2015–2018 in the pre-period would conflate two distinct policy shocks—the early trade-war skirmishes and the later sanctions regime—and so would complicate identification of the 2021 break. The 2019–2024 window provides a brief pre-period during which the early trade-war effects had settled and the major sanctions had not yet taken effect. Shock-timing clarity supplies the rationale for this restriction; identification is carried by the within-unit designs above, with the thin pre-period serving only as a descriptive check.

The Hong Kong cross-market extension affords an additional pre-trend check. Because board internationalization began later among Hong Kong-listed firms than among mainland A-share companies, the Hong Kong panel affords a longer pre-treatment window than the mainland data.

cannot match treated firms' pre-treatment trajectories from control units with near-zero baseline exposure.

The mainland and Hong Kong event studies appear in Appendix A.4 (Figures A1 and A2).

Our causal interpretation is bounded by where the design's identifying power lies, which is strongest for board composition, the outcome to which firms' pre-period U.S.-nexus exposure is mechanically and contemporaneously linked. We state that boundary once, as a ladder of claims that the results sections then take as given. The first rung is where the design's identifying power is strongest, and even there the claim is our most defensible causal interpretation rather than a settled effect: we treat the director-exit results, together with the placebo decomposition that isolates the U.S.-specific component from a concurrent market-wide de-internationalization, as the interpretation the evidence most credibly supports, while acknowledging the residual threats to identification we take up in Section 7. The second rung is descriptive: the board recomposition that follows, that is, who replaces the departing directors, which we characterize as an association conditional on the identified exit rather than as a separate causal effect, because replacement type is jointly determined with the firm's broader response to the shock. The third rung is suggestive: the downstream consequences for valuation, governance, and capital reallocation, which lie further from the source of variation because any effect of the shock on them would operate through several channels—direct sanctions exposure, customer and supplier disruption, financing conditions, and investor sentiment—that the exposure measure does not separate. We report each result at the rung its identification supports, and the results sections invoke this ladder rather than restate it. Figure 1 summarizes this logic and the underlying mechanism.

3.5 Summary Statistics

Table 1: Summary Statistics

Panel A: Firm-Year Variables (2019–2024)						
	N	Mean	SD	P25	Median	P75
U.S.-nexus director share	16,987	0.019	0.036	0.000	0.000	0.040
U.S.-nexus directors (count)	16,987	0.366	0.678	0.000	0.000	1.000
Foreign-director share	16,987	0.018	0.055	0.000	0.000	0.000
Board size	16,926	8.42	1.78	7.0	9.0	9.0
Independent-director ratio (%)	16,925	38.12	5.72	33.33	36.36	42.86
Log assets	16,987	22.73	1.575	21.62	22.48	23.57
Leverage	16,987	0.456	0.218	0.287	0.443	0.606
ROA	16,987	0.022	0.085	0.006	0.030	0.063
Top-3 executive pay (M¥)	16,901	4.35	4.60	2.10	3.10	4.96
Institutional ownership (%)	16,904	44.77	24.77	25.25	45.63	64.45
State-owned enterprise	16,987	0.323	—	—	—	—

Panel B: Treatment and Control Breakdown						
	Treated (1,040 firms)			Control (1,871 firms)		
	N	Mean	SD	N	Mean	SD
U.S.-nexus director share	6,056	0.049	0.044	10,931	0.003	0.013
Board size	6,034	8.56	1.91	10,892	8.33	1.69
Independent-director ratio (%)	6,034	38.04	5.78	10,891	38.16	5.69
Top-3 executive pay (M¥)	6,017	4.93	5.46	10,884	4.03	4.01
Institutional ownership (%)	6,030	45.25	24.61	10,874	44.51	24.86

Panel C: Sample Composition
2,911 unique firms (1,040 treated, 1,871 control); 16,987 firm-years
Pre (2019–2020): 5,637 firm-years; Post (2021–2024): 11,350 firm-years
SOE firms: ∼1,013 (5,480 firm-years); POE firms: ∼2,079 (11,507 firm-years)

Note: This table reports firm-year summary statistics for the preferred mainland A-share sample over 2019–2024 and the associated treatment-control breakdown.

Table 1 presents our summary statistics for the preferred sample of 16,987 firm-year observations spanning 2019–2024. Panel A reports firm-year-level variables. The average board has 8.4 directors (board of directors only, excluding the supervisory board and executives), with U.S.-nexus directors comprising 1.9% of the board on average. The mean independent-director ratio is 38.1%, just above the regulatory minimum of one-third, with 44% of firm-years bunched at exactly 33.3%. Top-3 executive compensation averages 4.35 million yuan, and aggregate institutional ownership is 44.8%, while state-owned enterprises comprise 32.3% of firm-years. Panel B breaks down key variables by treatment status: treated firms have higher average U.S.-nexus shares over the sample (4.9% vs. 0.3%; the sample-wide treated mean is depressed by the post-shock decline, and the 2019–2020 pre-shock treated mean is approximately 5.9%), slightly larger boards, and higher executive compensation.

4 The Exit of U.S.-Nexus Directors

We read the broad exit of U.S.-connected directors documented in this section as consistent with compliance chill, the channel through which the U.S. Persons Rule and the surrounding sanctions framework raised the personal legal and reputational cost of board service at politically exposed Chinese firms. The exit appears across firms facing no direct legal prohibition, concentrates among independent directors, and operates through passive non-renewal at term end, and each of these features maps onto the legal structure set out in Section 2. The concentration among independent directors follows from their fee- and equity-based compensation falling within the regulations' reach and from their lower switching costs; the appearance of the exit even at firms under no firm-specific prohibition follows from liability attaching personally to the U.S. person; and the clustering of departures at scheduled term end, with mid-term resignation playing little role, is the low-friction route by which a board sheds exposure once the legal cost of continued service rises. The exit itself sits on the causal rung of Section 3's ladder; the compliance-chill mechanism is our reading of why it occurs, the interpretation most consistent with the assembled evidence, though the data do not adjudicate it against alternatives such as reputational avoidance unrelated to legal liability or anticipatory firm-side board management.

4.1 The Within-Board Exit Hazard

Our primary identifying design, specified in Section 3, is the director-level exit hazard with firm $\times$ year fixed effects. The unit of analysis is the director-year, the outcome is whether a director departs the board before the following year, and the firm $\times$ year fixed effects restrict identification to the contrast between each U.S.-connected director and the non-U.S. colleagues serving on the same board in the same year. Because any firm-level trajectory, including the mean-reverting drift of a high-baseline firm, is differenced out by construction, a firm-level pretrend cannot generate the estimate.

Figure 3 plots the within-board exit gap by year as an event study around 2020, the omitted base year, whose enlarged hollow marker sits at zero by construction. Relative to that base, the gap is significantly negative in 2018 (-3.2pp, the opposite sign from everything that follows), statistically indistinguishable from zero at the 5% level in 2019 ($+2.0$pp, $p = 0.07$), and widens sharply once the prohibitions bind, reaching $+2.6$pp in 2021 (significant at the 5% level) and roughly $+4.6$pp in 2022 and $+4.3$pp in 2023 (each significant at the 1% level), against a mean annual exit rate near 14.5%. Estimated directly, the 2020 base-year gap is itself $+2.7$pp (significant at the 5% level), the year of the Entity List expansions and the signing of Executive Order 13959; because the 2019–20 cells already carry part of the response, pooled estimates that treat them as clean pre-years are conservative.

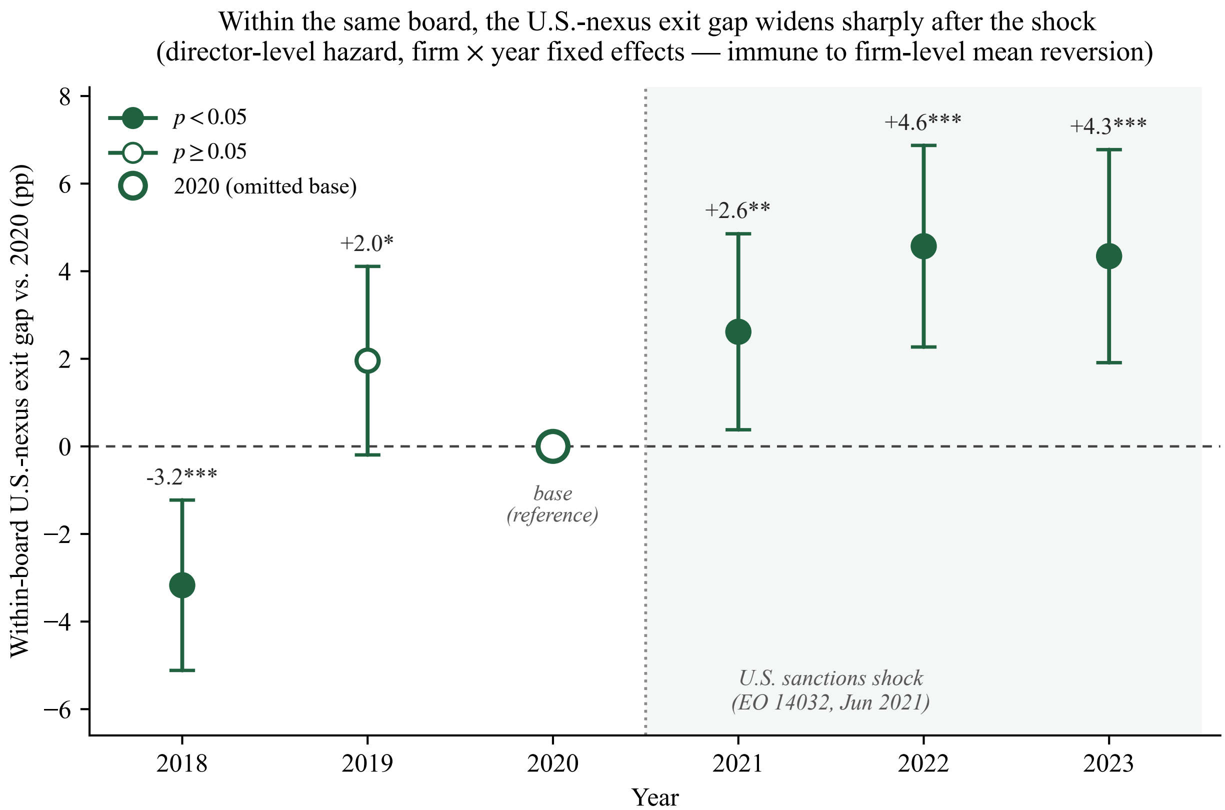

Figure 3: Within-board exit hazard for U.S.-nexus directors, by year, relative to the omitted 2020 base year. Each point is the coefficient on U.S.-nexus × year in a director-level linear-probability model of board exit with firm × year fixed effects, estimated on 319,979 director-years across 2,911 firms; whiskers are firm-clustered 95% confidence intervals (filled markers denote $p < 0.05$; the 2019 point, $+2.0$pp with $p = 0.075$, is significant only at the 10% level). The enlarged hollow marker at 2020 is the omitted base, plotted at zero by construction as a reference rather than an estimate. Because the fixed effects absorb every firm-level confounder, each coefficient compares U.S.-connected directors only with their non-U.S. colleagues on the same board in the same year, so a firm-level pre-trend or mean-reverting drift cannot generate it. The gap widens sharply after the prohibitions bind; the body text and Table 2 report the year-by-year and pooled estimates.

Table 2 reports the underlying estimates. Column 1 is a pooled linear-probability model; column 2 adds firm and year fixed effects; columns 3–5 absorb firm $\times$ year fixed effects, which restrict identification to the within-board contrast and difference out any firm-level mean reversion. The coefficient on U.S.-nexus $\times$ Post is positive in every specification, reaching $+3.1$pp (significant at the 1% level) over the full 2018–2023 window in column 4 and $+4.0$pp (significant at the 1% level) within treated firms in column 5; it attenuates to $+1.4$pp (not significant at the 10% level) only in the narrow 2019–2023 window, where the pooled interaction is diluted because the pre-period cells already carry an elevated gap ($+2.0$pp in 2019 and $+2.7$pp in 2020, consistent with pressure building through the trade-war escalation), so the full-window and treated-only columns are the informative pooled summaries. The positive U.S.-nexus main effect measures the pre-period within-board exit gap that Figure 3 absorbs into its 2020 base year.

Table 2: Director-Level Exit Hazard with Firm $\times$ Year Fixed Effects

	(1) Pooled	(2) Firm + Year FE	(3) Firm $\times$ Year FE	(4) Firm $\times$ Year (2018–23)	(5) Firm $\times$ Year (treated)
U.S.-nexus $\times$ Post	+0.0115	+0.0117	+0.0145	+0.0313***	+0.0399***
	(0.281)	(0.275)	(0.172)	(0.001)	(<0.001)
U.S.-nexus	+0.0227***	+0.0266***	+0.0236***	+0.0068	+0.0236***
	(0.003)	(0.001)	(0.002)	(0.251)	(0.002)
Firm FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Firm $\times$ Year FE	No	No	Yes	Yes	Yes
Window	2019–23	2019–23	2019–23	2018–23	2019–23
Sample	All	All	All	All	Treated
Mean exit rate	0.145	0.145	0.145	0.144	0.149
Director-years	270,116	270,116	270,116	319,979	99,125
Firms (clusters)	2,902	2,902	2,902	2,911	1,038

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The dependent variable is an indicator that a director leaves the board between year t and $t+1$; director-years whose firm is not listed in $t+1$ are dropped so that delistings are not counted as exits. Post $= 1$ for $t \geq 2021$. Columns 3–5 absorb firm $\times$ year fixed effects, so identification rests on the within-board contrast between U.S.-nexus and non-U.S. directors; column 4 uses the full 2018–2023 window and column 5 restricts the sample to treated firms. Standard errors are clustered by firm.

Scheduled term cycles are an unlikely explanation for the gap, since routine turnover falls on a director's U.S. and non-U.S. board colleagues alike and is absorbed by the fixed effects. Nor does it reflect U.S.-connected directors sitting systematically closer to term expiry than their boardmates: adding a director-tenure control leaves the post-shock widening intact (Appendix Table A8).

The timing of the departures follows the legal calendar. Because the changes file records an exact departure date for 96.9% of the exits we observe, each departure can be assigned to the legal era in which it occurred rather than to a roster year. Dated this way, U.S.-connected directors left at the same rate as their boardmates before the first investment prohibition took effect on January 11, 2021 ($+0.7$pp, $p = 0.22$); the within-board gap opens to $+2.1$pp during the investment-sanctions era and reaches $+2.8$pp under the Persons Rule (each significant at the 1% level), and among recorded U.S. nationals—the directors the rule binds personally—the Persons-Rule-era gap is the largest of any era, at $+4.1$pp (Appendix Table A9). A cohort view without survivor conditioning shows the cumulative toll: of the U.S.-connected directors serving on 2019 rosters, 61.8% had left those boards by 2023, against 52.8% of their own boardmates, a within-firm gap of 11.2 percentage points (Appendix Table A10). Section 7 draws on both tables in addressing the overlap with China's zero-COVID border controls.

4.2 The Firm-Level Exit

Table 3: Difference-in-Differences: U.S.-Nexus Director Share

	(1) Preferred 2019–2024 Post≥2021	(2) Original 2018–2024 Post≥2022	(3) Early Post 2019–2024 Post≥2020	(4) Extended 2018–2024 Post≥2021
DV: U.S.-nexus director share				
Treated × Post	-0.0209^{***}	-0.0209^{***}	-0.0167^{***}	-0.0188^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Alternative DVs (Column 1 specification only)				
DV: U.S.-nexus directors (count)	$\hat{\beta} = -0.3860^{***}$, $p < 0.001$			
DV: Any U.S.-nexus director (0/1)	$\hat{\beta} = -0.3507^{***}$, $p < 0.001$			
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Observations	16,987	19,600	16,987	19,600
R^2 (within)	0.0669	0.0664	0.0260	0.0523

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The main panel estimates difference-in-differences effects on the U.S.-nexus director share across sample windows, with controls, firm fixed effects, year fixed effects, and firm-clustered inference.

Table 3 reports our base difference-in-differences estimates of the geopolitical shock on U.S.-nexus board representation. In our preferred specification (Column 1, the 2019–2024 sample with Post $\geq$ 2021), the U.S.-nexus $\times$ Post coefficient is $\hat{\beta} = -0.021$ and is statistically significant at the 1% level, which indicates that treated firms experienced a 2.1 percentage-point decline in U.S.-nexus director share relative to control firms. Read against a treated-firm pre-shock mean of approximately 5.9%, this amounts to a one-third reduction. The estimate is notably stable across sample windows. Extending the panel to 2018–2024 in Column 2, which also shifts the post cutoff to 2022, yields an identical point estimate; holding the 2021 cutoff fixed over that window leaves it nearly unchanged (Column 4, -0.019^{***}); and moving the post cutoff earlier to 2020 in Column 3 produces a somewhat smaller but still highly significant effect (-0.017^{***}). This grid of cutoffs makes the dating-robustness point explicit: the instruments arrived continuously between November 2020 and October 2022, so no single post date is privileged, and the pooled estimate is insensitive to which one is chosen. Under industry $\times$ year fixed effects the estimate is virtually unchanged (-0.021^{***}), which confirms that the result does not reflect sector-wide trends. Even under two-way firm and year clustering, the most demanding standard-error specification we report, the coefficient remains significant at the 1% level despite a fourfold increase in the standard error.

Table 4: Event Study: Year-by-Treatment Interactions

	(1) Full Sample 2018–2024 Base = 2020	(2) Preferred 2019–2024 Base = 2019
DV: U.S.-nexus director share		
Treated $\times$ 2018	-0.0063^{***}	
	(<0.001)	
Treated $\times$ 2019	$+0.0002$	
	(0.836)	
Treated $\times$ 2020		-0.0003
		(0.804)
Treated $\times$ 2021	-0.0091^{***}	-0.0094^{***}
	(<0.001)	(<0.001)
Treated $\times$ 2022	-0.0169^{***}	-0.0172^{***}
	(<0.001)	(<0.001)
Treated $\times$ 2023	-0.0254^{***}	-0.0258^{***}
	(<0.001)	(<0.001)
Treated $\times$ 2024	-0.0322^{***}	-0.0325^{***}
	(<0.001)	(<0.001)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Clustering	Firm	Firm
Observations	19,600	16,987

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports year-by-treatment interactions for the U.S.-nexus director share, with base years and sample windows shown in the column headers.

Table 4 presents our event-study coefficients. In the preferred 2019–2024 specification (Column 2, with base year 2019), the single available pre-treatment coefficient, that for 2020, is -0.0003 and not significant at the 10% level, showing no detectable pre-shock divergence in the one pre-treatment year the window affords. The post-shock coefficients decline monotonically thereafter: -0.009^{***} (2021), -0.017^{***} (2022), -0.026^{***} (2023), and -0.032^{***} (2024). The full 2018–2024 specification (Column 1, with base year 2020) reveals a significant 2018 coefficient (-0.0063^{***}) alongside a clean 2019 reading ($+0.0002$, not significant at the 10% level). We regard the 2018 anomaly as an isolated blip attributable to the U.S.–China trade war, since it is smaller than every post-treatment coefficient and, as we show in the Discussion, is concentrated in a small set of early-treated firms and reverses sign once they are excluded. We accordingly adopt the 2019–2024 window as our primary specification.

Table 5: Two-Break Difference-in-Differences: Investment-Sanctions versus Persons-Rule Era

	(1) U.S.-nexus share	(2) U.S.-nexus count
Treated × Investment-sanctions era (2021–22)	-0.0131^{***}	-0.2487^{***}
	(<0.001)	(<0.001)
Treated × Persons-Rule era (2023–24)	-0.0289^{***}	-0.5286^{***}
	(<0.001)	(<0.001)
Escalation (Persons-Rule − investment era)	-0.0158^{***}	-0.2799^{***}
	(<0.001)	(<0.001)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Window	2019–24	2019–24
Observations	16,978	16,978
Firms (clusters)	2,902	2,902

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. The specification replaces the single post indicator with two era indicators: rosters 2021–22, the years in which the investment prohibitions bound, and rosters 2023–24, the first full years under the Persons Rule. The escalation row is the difference between the two era coefficients, with its standard error taken from the reparameterized regression (Treated × post plus Treated × second era). Column 1 uses the U.S.-nexus director share and column 2 the U.S.-nexus director count; both columns include the controls of Section 3 and firm and year fixed effects over 2019–2024. Observations differ from Table 3 only because the estimator drops nine firms observed in a single sample year, which are absorbed by their firm fixed effects and contribute to no estimate (16,978 observations and 2,902 clusters versus 16,987 and 2,911).

Table 5 sharpens these dynamics into the two-track legal structure of Section 2. The pooled specification dates the bundled regime to its August 2021 operative break; here we instead estimate one coefficient for the rosters of 2021–22, when the investment prohibitions bound, and one for the rosters of 2023–24, the first full years under the conduct-based Persons Rule. The design still identifies the cumulative response to the bundle—the concession of Section 3 stands—and the two coefficients ask only whether that response deepened once the sharper track arrived. It did: treated firms lose 1.3 percentage points of U.S.-nexus director share relative to controls in the investment-sanctions era and 2.9 percentage points in the Persons-Rule era, an escalation of -1.6 percentage points that is itself significant at the 1% level, and the director-count column shows the same doubling (from -0.25 to -0.53 directors). An escalating legal-exposure mechanism predicts this profile, because the rule that arrived in October 2022 attaches to a U.S. person's work rather than to securities transactions. The leading rival readings predict the opposite: China's border reopening in January 2023 should have eased a pandemic-friction effect in the second era rather than doubled it, and mean reversion supplies no reason for a decline to accelerate in its third year.

The director-level hazard traces the same two steps in event time (Figure 3), though its era contrast is statistically softer than the firm-level one. The cohort evidence explains why: the investment-sanctions era removed the most exposed directors from the later risk sets (Appendix Table A10), which attenuates late-era hazard contrasts, while a departure flow that merely holds at four to five percentage points across successive years integrates to the near-doubling of the stock effect reported here.

Table 6: Dose-Response and Position-Type Concentration

Panel A: Extension 1a — Dose-Response by Position Type			
	(1) U.S.-nexus share (independent)	(2) U.S.-nexus share (non-independent)	
Intensity × Post	-0.7734^{***}	-0.1031^{***}	
	(<0.001)	(<0.001)	
Controls	Yes	Yes	
Firm FE	Yes	Yes	
Year FE	Yes	Yes	
Clustering	Firm	Firm	
Observations	16,987	16,987	
Panel B: Extension 1c — Position Type Split (Binary Treatment)			
	(1) U.S.-nexus share (independent)	(2) U.S.-nexus share (non-independent)	(3) Independent-director share
Treated × Post	-0.0523^{***}	-0.0058^{***}	+0.0015
	(<0.001)	(<0.001)	(0.701)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Clustering	Firm	Firm	Firm
Observations	16,987	16,987	10,695
R^2 (within)	0.0615	0.0124	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The panels compare dose-response and binary-treatment estimates by director position type; the Panel B independent-to-non-independent coefficient ratio is $9.0\times$.

Table 6 reports two extensions. Panel A presents the dose-response gradient by position type, which is 7.5× steeper for independent directors (-0.773^{***}) than for non-independent directors (-0.103^{***}). Panel B reports the binary difference-in-differences by position type: -0.052^{***} for independent directors against -0.006^{***} for non-independent directors, a 9× ratio. We read this concentration among independents as consistent with the U.S. Persons Rule bearing on advisory relationships and with independent directors facing lower switching costs. Notably, the board independence ratio itself is unaffected ($+0.002$, not significant at the 10% level), which confirms that firms replaced departing U.S.-nexus independents with domestic independents and maintained board structure.

Table 7: Resignation Reasons as Quasi-exogeneity Evidence

Panel A: Euphemistic Resignation Rates (2×2)				
	Pre (2019–20)	Post (2021–24)	Diff	
U.S.-nexus	34.5% (n=513)	28.3% (n=1,079)	−6.2 pp	
Non-U.S.-nexus	39.1% (n=35,337)	38.8% (n=82,507)	−0.3 pp	
DiD $= -6.0$ pp, $\chi^2 = 55.77$, $p = 0.0000$				
Panel B: Linear probability model (DV: euphemistic resignation)				
	(1) Raw OLS	(2) Year FE	(3) Firm + Year FE	(4) Triple-Diff
U.S.-nexus × Post	−0.0758***	−0.0763***	−0.0818***	
	(0.004)	(0.004)	(0.003)	
U.S.-nexus × Post × Treated				−0.1479***
				(<0.001)
U.S.-nexus × Post				+0.0536
				(0.134)
U.S.-nexus × Treated				−0.1226***
				(<0.001)
Year FE	No	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Observations	75,674	75,674	75,674	75,674
Panel C: Decomposition by stated reason and position (Year FE + controls)				
	β(U.S.-nexus × Post)	p-value	DV mean	Obs.
Personal reasons	−0.0650***	(0.006)	15.0%	75,674
Term expiry	+0.0821***	(0.007)	35.0%	75,674
Euphemistic, independent directors	−0.1260***	(<0.001)	–	17,190

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports resignation-reason evidence using euphemistic resignation rates and linear probability models for U.S.-nexus departures before and after 2021.

Table 7 examines the stated reasons for resignation. Against the naive prediction that euphemistic reasons would increase, the post-shock U.S.-nexus departures show a decrease: $\hat{\beta}$(U.S.-nexus × Post) $= -0.076$ (significant at the 1% level). The triple interaction with Treated is stronger still (-0.148, significant at the 1% level). Decomposing the reasons reveals the mechanism (Panel C), since "personal reasons" declined (-0.065, significant at the 1% level) while "term expiry" increased ($+0.082$, significant at the 1% level), and the effect concentrates among independent directors (-0.126, significant at the 1% level). Taken together, these patterns indicate that U.S.-nexus exits occurred through passive non-renewal at term end, with firms waiting for natural board renewal cycles and forced mid-term resignations playing little role, which is precisely the departure pattern a chilling-cost mechanism predicts.

4.3 Board Reconstitution: Who Replaces the Departing Directors

In this section we characterize who fills the seats vacated by U.S.-nexus directors. Having interpreted the exit as the likely causal effect of the shock, we now turn to how boards recompose themselves around it. This board recomposition is the descriptive rung of Section 3's ladder, an association conditional on the identified exit rather than a separate causal effect.

Figure 4 sets the scene. At treated firms the U.S.-nexus director share falls by roughly a third after 2021, while non-U.S. foreign representation and overall board size hold steady, showing that the vacated seats are refilled; the remainder of this section asks who fills them.

Table 8: Replacement-Director Professionalization

Panel A: DV = Domestic technocrat (Replacement-Pair Level)				
	(1) Raw OLS	(2) Year FE	(3) Firm + Year FE	(4) Full
Treated (U.S.-nexus exit)	$+0.0262^{**}$	$+0.0378^{***}$	$+0.0095^{*}$	$+0.0095^{*}$
	(0.027)	(0.001)	(0.059)	(0.058)
Year FE	No	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Controls	No	No	No	Yes
Observations	310,645	310,645	310,645	310,645
R^2	0.0000	0.0023		
Panel B: DV = CCP-affiliated replacement (Preferred Specification)				
	(1)			
Treated (U.S.-nexus exit)	-0.0016			
	(0.706)			
Year FE	Yes			
Firm FE	Yes			
Controls	Yes			
Observations	310,645			

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table uses replacement-pair data to estimate domestic technocrat and CCP replacement outcomes for seats vacated by U.S.-nexus exits.

Table 8 examines the directors who step into the seats that departing U.S.-nexus directors leave behind. Drawing on 310,645 matched exit–entry replacement pairs, our preferred specification (Column 4) yields $\hat{\beta}(\text{Treated U.S. exit}) = +0.0095$ (significant at the 10% level), so that U.S.-nexus exits from treated firms post-shock are associated with a roughly one percentage-point higher rate of replacement by domestic technocrats. We read this association as a descriptive pattern in who fills the vacated seats. The CCP-replacement rate, by contrast, exhibits no differential change

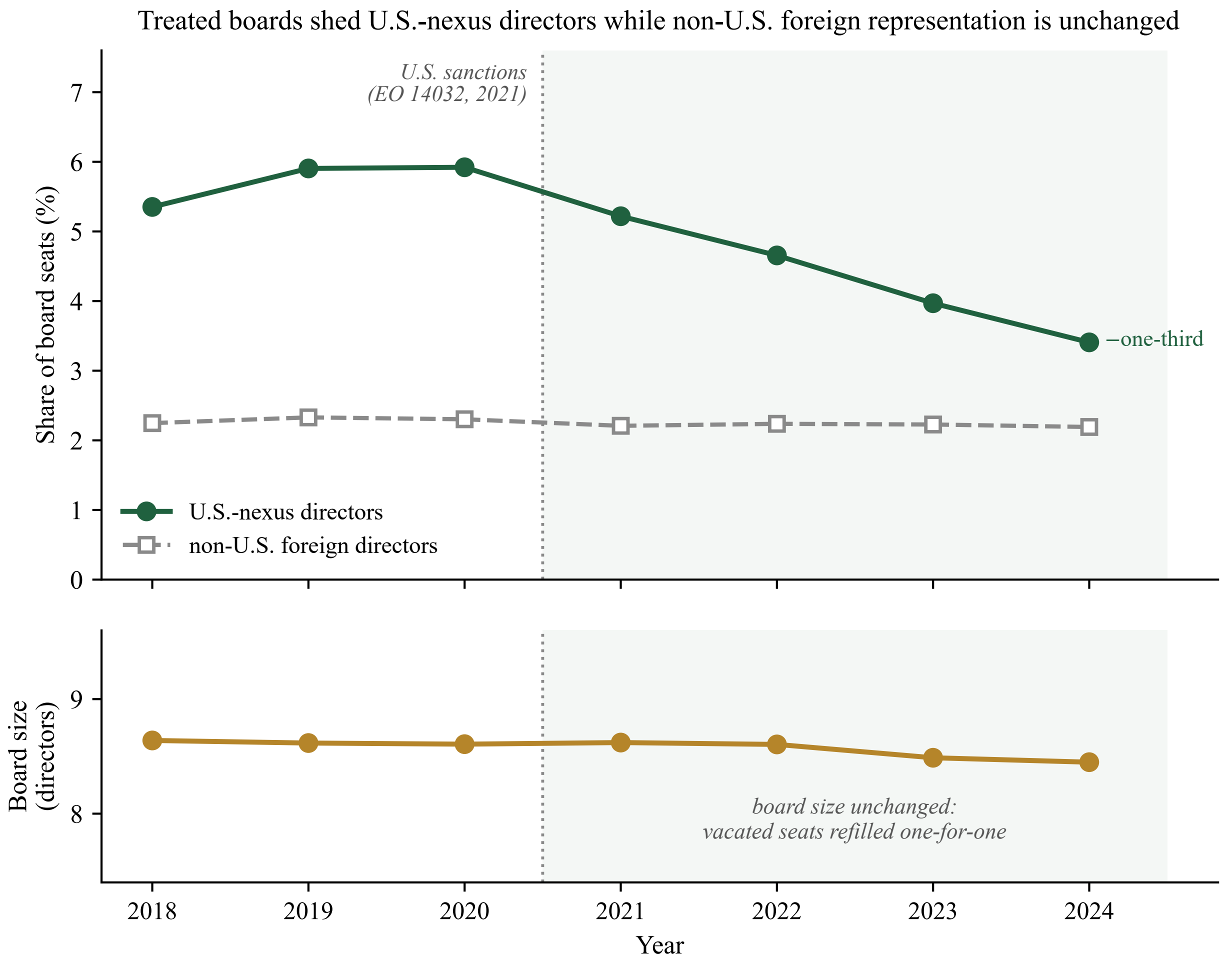

Figure 4: Board composition at treated mainland firms, 2018–2024. Treated firms are those with at least one U.S.-nexus director in the 2019–2020 baseline; means are unweighted across treated firms. Panel A plots the mean share of board seats held by U.S.-nexus directors and by non-U.S. foreign directors; Panel B plots mean board size (the CSMAR board-of-directors count). After the 2021 sanctions the U.S.-nexus share declines steadily, from roughly 5.9% at the 2019–2020 peak to roughly 3.4% by 2024, whereas non-U.S. foreign representation is flat—the U.S.-specific pattern that the within-firm placebo formalizes (Table 11). Board size is unchanged throughout, so the vacated seats are refilled one-for-one; the replacement analysis (Table 8) shows the replacements are disproportionately domestic technocrats.

(-0.002, not significant at the 10% level), a clean descriptive non-result that runs directly against a state-capture narrative.

Table 9 reports three extensions. Panel A shows that the composite technical-background index is higher for replacements of U.S.-nexus exits ($+0.007$, significant at the 1% level), a movement driven entirely by the professional-credential component ($+0.016$, significant at the 1% level). Panel B turns to board size: CSMAR's pre-computed board size (board of directors only, mean 8.4) registers no change ($+0.007$, not significant at the 10% level), consistent with complete one-for-one replacement. The previously reported shrinkage (-0.33, significant at the 5% level) was, notably, an artifact of the director-panel count, which erroneously included supervisory-board members and executives (mean 19.1). The CSMAR independent-director ratio is likewise unchanged (-0.098pp, not significant at the 10% level), which confirms that the independence ratio was maintained. Panel C examines compensation: independent-director pay at privately owned enterprises shows a positive but statistically insignificant movement ($+0.111$, not significant at the 10% level), so the earlier indication of upward pressure on independent-director pay does not survive the revised sample and we do not read a compensation response into it. Top executive compensation is null overall (-0.014, not significant at the 10% level), while SOE executive pay declined by about 4.5% (-0.045, significant at the 10% level), a different labor market that is likely shaped by administrative pay constraints. The Appendix consolidates robustness across all specifications, including alternative fixed-effect structures, clustering levels, and wild cluster bootstrap inference (Table A7).

The directors who depart are more highly credentialed than those who remain. Table 10 compares U.S.-connected directors with their non-U.S. colleagues on two credentials that lie outside the U.S.-nexus definition, measured in the 2019–2020 baseline. U.S.-connected directors hold a graduate degree at nearly twice the domestic rate (84.5% against 45.5%), a gap of 39 percentage points that survives when each U.S.-connected director is compared only with the non-U.S. colleagues on the same board in the same year ($+0.352$, significant at the 1% level). On senior domestic professional credentials the gap is smaller, and among independent directors—the group through which the exit runs—it narrows to a within-board difference that is not significant at the 10% level ($+0.029$). The departing directors thus carried distinctly more graduate-level and internationally oriented human capital, while the domestic professionals who remained and stepped into the vacated seats were their equals on domestic professional standing. This reading anticipates the consequence evidence of Section 6, where the monitoring outcomes show no detectable near-term deterioration while the valuation evidence suggests that the graduate-level international human capital that departed was priced at privately owned firms.

Table 9: Replacement: Credentials, Board Size, and Compensation

Panel A: Extension 2a — Technical-background index and Components

DV:	(1) Technical-background index	(2) Graduate degree	(3) Professional credential	(4) Domestic only	(5) Non-CCP
Treated (U.S.-nexus exit)	$+0.0069^{***}$ (0.004)	$+0.0052$ (0.367)	$+0.0161^{***}$ (<0.001)	$+0.0048$ (0.139)	$+0.0016$ (0.706)
Firm FE + Year FE + Controls	Yes	Yes	Yes	Yes	Yes
Observations	310,645	310,645	310,645	310,645	310,645

Panel B: Extension 2b — Board Size and Independence Ratio

	(1)	(2)	(3)
DV:	Board size	Independent-director ratio	Director count (incl. supervisors)
Source:	CSMAR comp/gov	CSMAR comp/gov	Director panel
Treated $\times$ Post	$+0.0074$ (0.856)	-0.0984 (0.566)	-0.3312^{**} (0.010)
Firm FE + Year FE + Controls	Yes	Yes	Yes
Observations	16,926	16,925	16,987

Panel C: Extension 2c — Compensation

	(1)	(2)	(3)	(4)	(5)
DV:	Log(indep. director pay)	Log(indep. director pay)	Log(indep. director pay)	Log(top-3 exec. pay)	Log(top-3 exec. pay)
Sample:	Full	POE only	SOE only	Full	SOE only
Treated $\times$ Post	$+0.0654$ (0.290)	$+0.1109$ (0.120)	-0.0449 (0.735)	-0.0137 (0.377)	-0.0454^{*} (0.069)
Firm FE + Year FE + Controls	Yes	Yes	Yes	Yes	Yes
Observations	7,319	5,334	1,985	16,888	5,454

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. Board size counts only the board of directors (mean 8.4); the director count also includes supervisors and executives (mean 19.1), so the apparent shrinkage in that count is an artifact. Independent-director pay is measured at the director level; top-three executive pay is measured at the firm level.

Table 10: Director Credentials: U.S.-Connected versus Domestic Directors

Panel A: All directors				
	(1) Domestic	(2) U.S.-nexus	(3) Gap (pp)	(4) Within-board
Graduate degree (Master's/PhD/MBA)	45.5	84.5	+39.0	$+0.352^{***}$
Senior professional title	33.7	46.4	+12.8	$+0.155^{***}$
Panel B: Independent directors only				
	(1) Domestic	(2) U.S.-nexus	(3) Gap (pp)	(4) Within-board
Graduate degree (Master's/PhD/MBA)	66.1	88.6	+22.5	$+0.217^{***}$
Senior professional title	54.0	56.6	+2.7	+0.029

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The sample is the mainland director panel in the 2019–2020 pre-shock baseline. Columns 1 and 2 report the share of directors holding each credential, in percent, for non-U.S. (domestic) and U.S.-nexus directors respectively; column 3 is the percentage-point gap; column 4 is the coefficient on U.S.-nexus in a linear-probability model of the credential indicator with firm $\times$ year fixed effects and standard errors clustered by firm, so that it compares each U.S.-connected director only with the non-U.S. colleagues on the same board in the same year. Graduate degree is a CSMAR education code of master's, doctorate, or MBA/EMBA; senior professional title flags senior-engineering, professorial, or accountancy (CPA) and equivalent designations in the CSMAR professional-title field. Neither credential enters the U.S.-nexus treatment definition. The within-board estimates are identified from the 1,786 baseline firm-years that contain both U.S.-connected and non-U.S. directors.

4.4 Isolating the U.S.-Specific Channel

A central question for our identification strategy is whether the director exit we document reflects U.S.-specific sanctions pressure or a broader de-internationalization of corporate boards. To address it, we report placebo designs across both markets in Table 11, and the resulting pattern furnishes the paper's most credible secondary check on identification. Under a counterfactual scenario in which market-wide de-internationalization drives the exit, one would expect non-U.S. foreign directors to leave treated firms alongside their U.S.-connected colleagues; the placebos test that prediction directly.

The within-firm placebo in columns 1–3 holds the firm fixed. At mainland treated firms, non-U.S. foreign directors show no detectable change ($+0.0005$, not significant at the 10% level), so that the elevated exit is confined to the U.S.-nexus group. At Hong Kong intent-to-treat firms (Hong Kong-listed Chinese firms with baseline U.S.-connected directors; Section 5 sets out that market's sample and treatment definitions), non-U.S. foreign directors increased ($+0.010$, significant at the 5% level), a movement consistent with substitution toward non-U.S. internationals. At the eleven explicitly sanctioned Hong Kong firms, non-U.S. foreign directors declined (-0.029, significant at the 10% level), which suggests that formal legal sanctions create a broader reputational deterrent reaching all foreign directors.

Table 11: Non-U.S. Foreign-Director Response: Cross-Market Placebo Tests

	(1) Mainland	(2) HK (ITT)	(3) HK (Sanctioned)	(4) HK (Placebo T2)
Panel A: US/Western Director Exit (Actual Effect)				
Treated × Post	-0.0209^{***}	-0.0131^{**}	-0.0135	-0.0001
	($<$0.001)	(0.012)	(0.309)	(0.894)
Treatment definition	U.S.-nexus	U.S.-nationality	Sanctioned	Other foreign
DV	U.S.-nexus share	U.S.-nationality share	U.S.-nationality share	U.S.-nationality share
Panel B: Non-US Foreign Directors at Treated Firms (Placebo)				
Treated × Post	$+0.0005$	$+0.0101^{**}$	-0.0294^{*}	-0.0236^{***}
	(0.651)	(0.034)	(0.053)	($<$0.001)
DV	Non-U.S. foreign share	Non-U.S. foreign share	Non-U.S. foreign share	Non-U.S. foreign share
Treated firms	1,040	137	11	519
Control firms	1,871	2,515	2,652	1,595
Observations	16,987	21,566	21,612	19,301
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. Columns 1–3 define treatment by baseline U.S.-nexus, U.S.-nationality, or sanctions exposure; column 4 defines placebo treatment as at least one non-U.S. foreign director in the baseline and zero U.S. directors. The Hong Kong specifications include the controls of Section 3; the mainland Panel B placebo is estimated with firm and year fixed effects and no controls.

Column 4 reports an unexposed-firm placebo. Firms that held non-U.S. foreign directors but no U.S. directors in the baseline experienced a marked decline in their own-type directors (-0.024, significant at the 1% level), revealing a concurrent market-wide de-internationalization in Hong Kong boards that operates independently of U.S. sanctions. The event study displays clean pre-trends in 2018–2019 and a monotonic decline after 2021, a profile that points to the National Security Law, COVID-era border restrictions, and the wider Sino-Western tensions as contributors to foreign-director attrition across the market. An analogous test on the mainland, among firms with non-U.S. foreign directors but no U.S.-nexus exposure, likewise yields a significant decline (-0.016, significant at the 1% level; Table A6).

Taken together, the cross-market pattern supports three conclusions. First, the evidence points to a genuine U.S.-specific channel, because at treated firms U.S. directors exit while non-U.S. foreign directors are unaffected on the mainland and increase in Hong Kong, a pattern that diverges from the de-internationalization counterfactual. Second, a distinct de-internationalization channel operates on different firms and through a different mechanism. Third, the surname-based estimate (-0.025^{***}) captures both channels, whereas the nationality-based estimate (-0.013^{**}) isolates the U.S.-specific component, and the gap between the two is consistent with the market-wide trend we observe at unexposed firms. Appendix A.4 reports robustness to alternative treatment definitions.

5 Cross-Market Replication: Hong Kong

The mainland results establish that exposed firms shed U.S.-nexus directors after the shock, with the within-board hazard supplying the causal anchor, and they document the domestic professionalization of the replacement directors that followed. A natural question is whether these effects generalize beyond the A-share universe. The Hong Kong Stock Exchange hosts a distinct population of Chinese firms, many of them internationally oriented, which are subject to different regulatory and disclosure regimes and which report quarterly ownership breakdowns that are unavailable in A-share data. In this section we extend the analysis to this independent sample to test whether the same shock produced parallel director exits and how board substitution differs across the two markets; the capital-base response, which surfaces only among the explicitly sanctioned firms, is reported in Appendix A.3.

5.1 Foreign-Director Exit in Hong Kong

Table 12: Foreign-Director Exit: Hong Kong-Listed Chinese Firms

	(1)	(2)	(3)	(4)
	DV: Western-foreign director share		DV: Foreign-dialect director share	
Panel A: Baseline DiD				
Treated × Post	-0.0274^{***}	-0.0253^{***}	-0.0260^{***}	-0.0224^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Panel B: Dose-Response				
Intensity × Post	-0.2695^{***}	-0.2449^{***}	-0.1199^{***}	-0.1092^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Controls	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Panel A statistics				
Observations	23,869	21,612	23,869	21,612
Firms	2,606	2,535	2,606	2,535
R^2 (within)	0.0450	0.0418	0.0214	0.0183
Panel B statistics				
R^2 (within)	0.0972	0.0884	0.0380	0.0349

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The panels report baseline difference-in-differences and dose-response estimates for Hong Kong-listed Chinese firms using Western-surname and foreign-dialect director-share outcomes.

Table 12 replicates the mainland exit result in an independent panel of Hong Kong-listed Chinese firms (roughly 2,600 firms over 2014–2025, with estimation samples that are somewhat smaller after controls and coverage restrictions, as reported in the table). We define treatment as having ≥ 1 Western-surname director in the 2019–2020 baseline, with Post ≥ 2021. Panel A reports the baseline DiD. The Western-surname measure yields $\hat{\beta} = -0.025$, an estimate that is significant at the 1% level, so that treated firms experienced a 2.5 pp decline in Western director share, a magnitude that closely parallels the mainland 2.1 pp figure. The broader foreign-dialect measure produces a similar effect (-0.022, significant at the 1% level), and Panel B confirms a dose-response gradient (-0.245, significant at the 1% level).

An independent nationality-based classifier decomposes this effect. Defining treatment as ≥ 1 U.S.-nationality director in the 2019–2020 baseline (137 firms, excluding the 11 sanctioned) and using the share of U.S.-nationality directors as the dependent variable, we obtain an intent-to-treat estimate of -0.013, which is significant at the 5% level (Table 11). The gap between the surname estimate (-0.025) and the nationality estimate (-0.013) is consistent with a sizable non-U.S. Western component of the exit, a pattern that fits a broader de-internationalization trend distinct from the U.S. sanctions; because the two classifiers rest on different treatments and largely distinct firm populations (Appendix A.1), we read them as triangulating replications rather than a formal decomposition. Because the nationality measure, like the mainland three-prong measure, is nationality-based, it provides the most direct comparison with the mainland figure.[12]

At intent-to-treat firms, the within-firm placebo of Section 4 showed the share of non-U.S. foreign directors rising post-shock (Table 11), a movement consistent with boards offsetting U.S.-connected departures by recruiting other international directors, although the design cannot confirm that the recruitment was a deliberate substitution response. This pattern stands apart from the mainland one, in which firms replaced U.S.-nexus directors with domestic technocrats. The divergence plausibly reflects differing labor-market depths, because the Hong Kong exchange hosts a deep pool of non-U.S. international professionals (the baseline non-U.S. foreign share is 5.6%), whereas mainland A-share boards draw from a domestic talent market.

At the eleven explicitly sanctioned firms, the pattern differs: non-U.S. foreign directors also declined despite carrying no direct U.S. compliance exposure (Table 11), a marginal estimate drawn from a small set of firms that warrants caution.

Read together, the two markets are consistent with a single mechanism operating at different intensities of legal exposure: under intent-to-treat exposure the response is selective, falling on U.S.-connected directors while sparing other foreign directors, whereas explicit sanctions produce a broader decline in foreign directors of all origins together with measurable capital reallocation. Because the indiscriminate pole rests substantially on the eleven explicitly sanctioned firms and on the capital-base evidence, we treat it as suggestive and develop it, alongside the full international-ownership results and the cross-market synthesis, in Appendix A.3.

6 Does the Loss of U.S.-Connected Monitors Matter?

Sections 4 and 5 establish that the shock removed internationally credentialed, disproportionately independent directors from exposed boards. This section asks what that removal cost, in two steps: we first test whether the monitoring function those directors performed deteriorated, across price-based and internal-governance outcomes, and we then ask whether the market priced the human capital that departed. Both steps run on the mainland panel; the Hong Kong capital-base response, a parallel margin of the same shock rather than a downstream consequence of director loss, appears in Appendix A.3. The answer is one of bounded resilience, and a synthesis closes the section by drawing the two steps together: the monitoring function survived the exits, while a valuation cost surfaced at the private firms least able to replace the departing

[12] All Hong Kong director-exit results are robust to controlling for log market capitalization (22 of 22 specifications unchanged). We omit this control from the baseline to avoid observation loss (6.4% missing) and a potential bad-control concern, as market valuations may themselves respond to the shock.

directors' international human capital.

6.1 Monitoring Outcomes

Whether the exit carries a governance cost is the question that gives the board-composition channel its economic stakes, because the departing directors are precisely the type that prior work associates with stronger monitoring and with constraints on controlling-shareholder expropriation in Chinese firms (Zhu et al., 2016; Gong et al., 2021; Jiang et al., 2010). To test that consequence directly, we take stock-price crash risk as our primary governance outcome. Crash risk is the most widely used market-based proxy for the consequences of weak board oversight: a large literature holds that when monitoring is weak, managers can hoard adverse information until it accumulates and is released in infrequent large drops, so that a return distribution more prone to sudden crashes is the price-based signature of the opacity that weakened oversight permits. The measure is well suited to our setting. Because it is computed from each firm's own weekly returns, it is available for the full panel rather than only for the firms and years in which a discrete governance event is observed, which gives it far broader coverage than an event-based test. It also aggregates the consequences of monitoring quality into a single outcome without requiring us to pre-specify which governance margin the departing directors disciplined, an advantage where the monitoring functions at issue are diffuse. In Chinese firms, moreover, crash risk has been tied directly to board independence and to directors with overseas experience (Cao et al., 2019), the attributes the sanctions shock stripped from exposed boards. A monitoring-driven deterioration of the kind we hypothesize should therefore register in this measure if it registers in prices anywhere. We construct the negative coefficient of skewness (NCSKEW) and the down-to-up volatility ratio (DUVOL), the two conventional crash-risk measures, for which higher values denote a return distribution more prone to sudden large declines, and regress each on the treated-by-post interaction with firm and year fixed effects, the controls of Section 3, and firm-clustered inference over 2019–2024.

We find no governance cost on this measure, and the confidence bounds are informative. As Table 13 reports, the treated-by-post coefficient is -0.009 for crash-risk skewness ($p = 0.73$) and -0.014 for down-to-up volatility ($p = 0.42$), each statistically indistinguishable from zero at conventional thresholds. The 95% confidence interval excludes effects beyond roughly one-tenth of a standard deviation in either direction, and the minimum detectable effect at 80% power is about 0.10 standard deviations, so a governance-relevant deterioration would have been detected had it occurred. The null is unmoved by pooled estimation, a lagged dependent variable, two-way clustering, the broad-market benchmark that adds the ChiNext and STAR boards, a thirty-week minimum, and dose-response specifications that scale treatment by baseline U.S.-nexus exposure or by the realized post-shock loss of U.S.-connected directors; the year-by-year event study is likewise flat, with no post-2021 break.

Table 13: Crash Risk and the Loss of U.S.-Nexus Directors

	Crash risk (neg. skewness) (1)	Crash risk (down-to-up vol.) (2)
Treated $\times$ Post (primary)	-0.009 (0.731)	-0.014 (0.423)
95% confidence interval	[-0.063, $+0.044$]	[-0.047, $+0.020$]
Robustness, Treated $\times$ Post:		
Pooled OLS	-0.009 (0.743)	-0.013 (0.437)
Lagged dependent variable	-0.004 (0.890)	-0.013 (0.472)
Two-way cluster (firm, year)	-0.009 (0.682)	-0.014 (0.423)
Broad-market benchmark	-0.008 (0.763)	-0.012 (0.488)
Minimum 30 weeks	-0.008 (0.755)	-0.013 (0.419)
Dose-response (per unit), $\times$ Post:		
Baseline U.S.-nexus exposure	$+0.172$ (0.587)	-0.027 (0.895)
Realized loss (count)	-0.034 (0.345)	-0.026 (0.269)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Controls	Yes	Yes
Observations (firm-years)	16,639	16,639
Firms	2,880	2,880
Min. detectable effect (80%)	$\approx$0.10 SD	$\approx$0.10 SD

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The dependent variables are firm-year stock-price crash-risk measures built from firm-specific weekly returns: NCSKEW, the negative coefficient of return skewness, and DUVOL, the log ratio of return volatility in below-average (“down”) weeks to above-average (“up”) weeks; for both, higher values indicate greater crash risk. Treated $\times$ Post is the exposure difference-in-differences interaction (Post $=$ 1 for years $\geq$ 2021); the primary specification adds firm and year fixed effects, the controls listed in Section 3, and firm-clustered standard errors over 2019–2024. The robustness rows vary that specification one change at a time; the dose-response rows replace the binary treatment with continuous baseline exposure or with the realized count of U.S.-connected directors lost, each interacted with Post, so their per-unit scale differs. The market benchmark is the Shanghai-plus-Shenzhen main-board A-share market excluding the ChiNext and STAR boards, and the broad-market row adds those boards. The minimum detectable effect is computed at 80% power against the estimated standard error. Crash-risk firm-years require at least 26 valid trading weeks.

Crash risk is the best price-based proxy available to us, yet some monitoring functions may move governance margins before they move prices. We therefore complement it with two internal-governance outcomes that bear more directly on the monitoring and minority-protection channels: the sensitivity of executive pay to firm performance and payout policy. Each is estimated as a reduced-form exposure difference-in-differences with firm and year fixed effects and firm-clustered inference over 2019–2024 on the non-financial sample, and we read the estimates as intent-to-treat consequences of baseline U.S.-nexus exposure rather than as the isolated causal effect of director loss.

Incentive alignment is the first channel. If internationally credentialed monitors disciplined the link between pay and performance, their departure should weaken it. We estimate a triple difference in which log executive compensation is interacted with firm performance, treatment, and the post indicator, so that the coefficient on the triple interaction measures the change in pay-performance sensitivity at exposed firms after the shock. Executive pay tracks performance strongly in levels—a one-unit increase in earnings per share raises top-three executive compensation by 0.144 log points (significant at the 1% level)—but the treated-by-post slope shift is economically negligible and statistically indistinguishable from zero (-0.012, $p = 0.54$; Table 14). The null is unchanged when performance is standardized within industry-year cells, when a quadratic performance term is added, under industry-by-year fixed effects, with the all-personnel compensation caliber, and over the extended 2018–2024 window. Pay-performance sensitivity does fall market-wide after 2021 (the performance-by-post coefficient is -0.096, significant at the 1% level), but this common decline falls equally on treated and control firms and is unrelated to de-internationalization.

Table 14: Pay-Performance Sensitivity and the Loss of U.S.-Nexus Directors

	(1)	(2)	(3)	(4)
Dep. var.:	ln Top-3 pay	ln Top-3 pay	ln Top-3 pay	ln Total pay
Performance:	EPS	ROA	Stock return	EPS
Performance (base slope)	0.144***	0.957***	0.093***	0.131***
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Perf. × Treated × Post	-0.012	-0.125	-0.026	-0.023
	(0.544)	(0.566)	(0.471)	(0.299)
Perf. × Treated	0.007	-0.095	0.007	0.014
	(0.749)	(0.612)	(0.824)	(0.561)
Perf. × Post	-0.096***	-0.879***	-0.055***	-0.085***
	(<0.001)	(<0.001)	(0.005)	(<0.001)
Observations	16,404	16,406	16,037	16,404
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. The dependent variable is log executive compensation (top-three senior managers in columns 1–3; all senior managers in column 4). Each column is a triple difference of pay on the indicated performance measure (basic EPS, ROA, or annual stock return), interacted with the baseline treatment indicator and the post indicator. The row of interest, Performance × Treated × Post, is the change in pay-performance sensitivity at exposed firms after 2021; the base slope reports the cross-sectional pay-performance gradient. All columns include firm and year fixed effects and the controls of Section 3 (firm size and leverage; profitability is excluded as it is the channel). Sample: non-financial firms, 2019–2024.

Payout policy is the second channel. Under the outcome model of dividends, weaker minority-shareholder protection predicts lower payouts, as controlling shareholders retain or divert cash rather than distribute it. We find no such movement. The treated-by-post effect is null for dividends scaled by assets ($+0.001$, $p = 0.30$), for the probability of paying a dividend ($+0.014$, $p = 0.29$), for the payout ratio among profitable firm-years (-0.000, $p = 0.99$), and for post-tax dividends per share (Table 15); a continuous dose-response in baseline exposure is likewise null, and the result is unchanged under industry-by-year fixed effects and a conditional specification that adds the profitability control. The estimates are informative rather than merely imprecise, since the minimum detectable effect on the payout ratio is about four percentage points against a sample mean near thirty percent, so a payout contraction of governance-relevant magnitude would have been detected had it occurred.

Table 15: Dividend Policy and the Loss of U.S.-Nexus Directors

	(1) Dividends/assets	(2) Pays dividend	(3) Payout ratio	(4) DPS (post-tax)
Panel A. Firm and year fixed effects (unconditional)				
Treated × Post	0.001	0.014	-0.000	-0.003
	(0.299)	(0.290)	(0.985)	(0.751)
Panel B. Adds ROA control (conditional payout policy)				
Treated × Post	0.000	0.005	0.004	-0.009
	(0.498)	(0.720)	(0.798)	(0.277)
Observations	16,450	16,450	13,162	16,450
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. The dependent variables are total cash dividends scaled by assets, an indicator for paying any cash dividend, the payout ratio (cash dividends over parent net income, defined for profitable firm-years), and post-tax dividends per share (zero-filled for non-payers). Treated × Post is the exposure difference-in-differences interaction (Post $= 1$ for years $\geq$ 2021). Panel A is the unconditional specification with firm and year fixed effects; Panel B adds the profitability (ROA) control and is read as conditional payout policy. Sample: non-financial firms, 2019–2024.

A further consequence test, reported in the appendix for completeness, points in the same direction. It asks how the market priced the departures themselves: in an event study of U.S.-connected director departures, cumulative abnormal returns around the departure date are economically small and statistically indistinguishable from zero across a battery of parametric and nonparametric tests, so the market registered no robust reaction to the loss of these directors (Appendix A.3, Table A5).

6.2 Valuation

A separate question is whether the human capital that departed was priced. The directors who left carried distinctly more graduate-level and internationally oriented credentials than the colleagues who remained and the domestic professionals who replaced them (Section 4, Table 10). One-for-one substitution preserved the board's domestic professional standing, and the monitoring outcomes above suggest that the oversight function survived the swap. What a domestic replacement cannot readily supply is certification and network capital of the departing kind: the credibility that a U.S.-trained or internationally licensed director lends to disclosures, cross-border transactions, and relationships with foreign counterparties. If investors valued that capital, its loss should register in Tobin's Q even while governance metrics hold; directors with foreign experience have indeed been shown to raise the performance of Chinese firms through channels of exactly this kind (Giannetti et al., 2015). The prediction is sharpest at privately owned firms, and any test that turns on the education of the departing director must be run within them. A highly educated U.S.-connected director plausibly wields less influence at a state-owned issuer than at a privately owned one: state firms draw on financing access and implicit guarantees that blunt the marginal value of any individual director's standing, and their formalized Party-committee structures place major decisions upstream of the board (Lin and Milhaupt, 2021). Where an individual director carries little decision weight, the education of the one who departs should matter little for value, so the education moderation below is estimated on the private-firm sample.

A first look uses the paper's baseline design unchanged: the same treatment indicator, the same Post ≥ 2021 window, and the same fixed effects and controls as the exit regressions of Section 4, with Tobin's Q as the outcome. Table 16 reports the estimates. Treated firms lose 0.089 of Tobin's Q relative to control firms after the shock ($p = 0.006$). Splitting the panel at the firm level locates the decline among privately owned firms, where the effect is -0.111 ($p = 0.012$), while the state-owned estimate is smaller and statistically indistinguishable from zero (-0.065, $p = 0.131$). The formal moderation test points in the same direction without settling the comparison, since the triple-difference offset for state-owned firms is positive but imprecise ($+0.048$, $p = 0.428$); we accordingly read the concentration at private firms from the subsample contrast rather than from the interaction.

Table 16: Tobin's Q and the Loss of U.S.-Nexus Directors, by Ownership

	(1) All firms	(2) POE only	(3) SOE only	(4) Triple difference
Treated × Post	-0.089^{***}	-0.111^{**}	-0.065	-0.111^{**}
	(0.006)	(0.012)	(0.131)	(0.013)
Treated × Post × SOE				$+0.048$
				(0.428)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	16,978	10,920	6,058	16,978

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. The dependent variable is Tobin's Q, winsorized at the 1st and 99th percentiles. Treated × Post is the exposure difference-in-differences interaction of Section 3 (Post $= 1$ for years ≥ 2021); all columns include firm and year fixed effects and the controls of Section 3 over 2019–2024. Columns 2 and 3 split the panel at the firm level into privately owned (never state-owned across the firm's panel years) and state-owned (ever state-owned) firms; column 4 estimates the pooled specification with the treatment effect interacted with the same firm-level state-ownership indicator.

The first look treats all exposed firms alike; to tie the valuation decline to the specific directors who left, we exploit variation in who departed. For each privately owned treated firm we date treatment from its first U.S.-nexus director exit in the panel, in whatever year it occurs, and ask whether the post-exit path of Tobin's Q varies with the education of the exiting director. The design is a cohort-stacked difference-in-differences: each exit cohort is paired with the never-treated private firms in a window from two years before to two years after the exit, the year before the exit serves as the reference period, and the stacked regression carries stack-by-firm and stack-by-year fixed effects with standard errors clustered by the original firm. The estimation sample comprises 1,897 private firms, among which 568 treated firms have observed education for the exiting director.

Table 17 reports the results. The post-exit change in Tobin's Q is 0.336 lower when the exiting director held a postgraduate degree ($p = 0.070$); in levels, firms losing a postgraduate-credentialed director decline (-0.109, $p = 0.024$) while firms losing a less-educated director do not ($+0.227$, $p = 0.207$). The event-time path behaves as the interpretation requires. The two groups are statistically indistinguishable before the exit ($+0.281$ at $e = -2$, $p = 0.152$), diverge negligibly in the exit year, and separate afterward, with the largest gaps one and two years out (-0.398 and -0.424, with $p = 0.112$ and $p = 0.110$). The ordered-rank moderator points the same way with less precision (-0.108, $p = 0.161$). A heterogeneity-robust cross-check is consistent: a Callaway and Sant'Anna (2021) estimator comparing treated with never-treated firms puts the average post-exit effect at -0.083 within private firms ($p = 0.087$) and at -0.089 for the informative-exit subset of departures ($p = 0.107$).

Table 17: Tobin's Q and the Education of Exiting U.S.-Nexus Directors

	Pooled post	Event time			
	(1)	(2)	(3)	(4)	(5)
	Post	$e = -2$	$e = 0$	$e = +1$	$e = +2$
Postgraduate interaction	-0.336^{*}	$+0.281$	$+0.026$	-0.398	-0.424
	(0.070)	(0.152)	(0.869)	(0.112)	(0.110)
Ordered education-rank interaction	-0.108	$+0.100$	$+0.026$	-0.141	-0.123
	(0.161)	(0.205)	(0.701)	(0.155)	(0.250)
Treated POE firms with known education	568				
POE firms	1,897				

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by the original firm. The sample is privately owned (never state-owned) firms; treatment is the firm's first U.S.-nexus director exit in the panel, in whatever year it occurs. Each exit cohort is stacked with the never-treated private firms over event years $e = -2$ to $e = +2$, with $e = -1$ omitted as the reference year, and the regression includes stack-by-firm and stack-by-year fixed effects. Each reported coefficient is the interaction of the indicated event-time (or pooled post) indicator with the education moderator: an indicator for a postgraduate degree (master's, MBA/EMBA, or doctorate) of the exiting director, or an ordered education rank (0 = undergraduate or below, 1 = master's/MBA/EMBA, 2 = doctorate). Where several directors exit in the firm's first exit year, the firm is coded by the highest education among them. Education is observed for 568 of the treated private firms.

Three qualifications bound this evidence, and we advance it as suggestive rather than established. Education is observed only for directors who exit, so the moderation is identified within treated firms, and the treated group is heavily imbalanced toward postgraduate exits. The pooled interaction clears only the ten-percent threshold. The result is also design-dependent, in that it appears in the private-firm stacked specification with the postgraduate moderator while cross-sectional comparisons among treated exits and moderation tests on the full panel are null. We read the pattern as consistent with a brain-loss channel, in which the market prices the loss of scarce, internationally portable human capital at the firms least able to substitute for it, and we do not claim more.

6.3 Synthesis: Bounded Resilience

Read together, the two halves of this section tell one story. Exposed firms shed internationally credentialed monitors without a detectable near-term cost on any monitoring margin we observe, as crash risk, the market's reaction at departure, pay-performance sensitivity, and payout all return bounded nulls. The most natural reading ties those nulls to the one-for-one domestic substitution documented in Section 4: vacated seats were refilled by professionals who matched the departing directors on domestic professional standing (Table 10), so no governance vacuum opened, and the firm's opacity profile need not move. What the replacements could not supply is the departing directors' international certification and network capital, and the valuation evidence suggests the market priced that gap at the firms for which it binds, since Tobin's Q declines at privately owned treated firms, most clearly after the exit of postgraduate-credentialed directors, while the state sector shows no comparable movement. The policy reading is pointed: a sanctions regime aimed at state-linked entities imposed its one measurable cost on private firms, whose access to internationally credentialed directors is a market relationship rather than a state guarantee. These consequence findings sit on the suggestive rung of Section 3's ladder. The director exit is plausibly identified by the external sanctions shock, whereas the links from board composition to crash risk, to the departure reaction, and to valuation run through many confounding influences—contemporaneous performance, disclosure choices, and market conditions—that the consequence designs do not separate; the valuation result in particular is design-dependent and clears only the ten-percent threshold in its pooled form. Because price-based measures cannot speak to longer-horizon or harder-to-observe costs, we read the monitoring evidence as bounded resilience rather than as a clean null. The channels that remain least visible to prices—enforcement actions, financial restatements, and tunneling through abnormal other-receivables—require enforcement and balance-sheet data still being assembled, which we leave as future evidence (Appendix A.3).

7 Limitations

With the cross-market evidence in place, this Article sets out the qualifications that bound its interpretation, taking in sequence the threats to identification, the caveats that attach to our measures, and the extensions that would tighten the design. Identification is carried by the within-firm and within-board comparisons of Section 3; the quasi-exogenous timing of the U.S. sanctions—the Entity List expansions, Executive Order 14032, and the U.S. Persons Rule—motivates the post-period indicator and the estimation window, while the causal claim rests on the within-unit comparisons. In our preferred 2019–2024 specification the single available pre-treatment coefficient (2020, base year 2019) shows no detectable divergence (-0.0003, not significant at the 10% level), although, as Section 3 explains, the design derives its identifying leverage from the within-firm and within-board comparisons. The extended 2018–2024 specification returns a 2018 coefficient that is negative and significant at the 1% level (-0.0063), a pattern we trace to 124 "early-treated" sample firms whose U.S.-nexus shares declined steeply before 2019, coincident with the Section 301 trade-war tariffs of July 2018 (a data-driven set—the treated firms with the steepest pre-2019 declines—since no Entity-List crosswalk is available); once these firms are excluded the negative 2018 coefficient reverses to $+0.004$ (significant at the 1% level), confirming that the anomaly is concentrated in that group. We therefore adopt the 2019–2024 window as our baseline. Were the sanction dates to arrive at genuinely different times across firms, a heterogeneity-robust estimator in the manner of Callaway and Sant'Anna (2021) would be the appropriate response; the explicit sanctions data are in any event too thin to support such standalone analysis—of the firms named on the U.S. lists, 27 appear in the panel and only 6 overlap with the treatment group—a scarcity that itself vindicates our intent-to-treat approach.

Because our treatment bundles several legal instruments together with other events of the 2020–2021 period—the pandemic, the broader trade conflict, and, in Hong Kong, the National Security Law—a natural concern is that one of these contemporaneous shocks produced the exit. The within-board design speaks to this concern as squarely as it does to mean reversion. Any shock that operates at the level of the firm, depressing all board service or all foreign board service at an exposed firm, is absorbed by the firm $\times$ year fixed effects and cannot generate the estimated gap. To survive the within-board comparison, a confounder would have to drive out a firm's U.S.-connected directors differentially relative to their own non-U.S. colleagues, and to do so on the timing of the 2021 sanctions. The pandemic, the trade war, and the National Security Law carry no such U.S.-specific, within-board signature, whereas the U.S. Persons Rule carries one by construction. The within-firm placebo reinforces this reading, since at mainland treated firms the elevated exit is confined to U.S.-connected directors while non-U.S. foreign directors are unaffected (Table 11), which isolates the U.S.-specific component of the decoupling bundle from the concurrent market-wide de-internationalization.

The pandemic warrants a further, timing-based answer, because zero-COVID border controls (March 2020 through the reopening of January 8, 2023) overlap the sanctions window and might be thought to bear hardest on internationally connected directors. If border friction drove the departures, the within-board gap should revert once the border reopened; it does not. The 2023 gap—estimated entirely on post-reopening departures—is $+4.3$pp (one-sided $p < 0.001$ against reversion) and statistically indistinguishable from its 2022 peak ($p = 0.90$; Appendix Table A9). Cutting finer with exact departure dates, the gap within the Persons-Rule era is present in the final zero-COVID months and larger in the reopened months ($+0.6$pp against $+2.1$pp), the U.S.-connected share of departures is essentially flat across the reopening date, and among recorded U.S. nationals—the group most exposed to border friction—the reopened months alone carry a gap of $+3.5$pp (significant at the 5% level). The era ladder that Section 4.1 reports from the same table makes the affirmative case, rising from a pre-prohibition null to its Persons-Rule peak, and the firm-level stock effect doubles between the two eras (Table 5) precisely when reopening should have eased a friction effect.[13] We accordingly read the departure years in three regimes: through 2019, a pre-pandemic and pre-sanctions baseline; 2020 through 2022, a band in which zero-COVID, anticipation, and the investment prohibitions overlap, and where our claim is the U.S.-specificity established by the placebo rather than clean timing; and 2023, post-reopening and inside the Persons-Rule regime, where the no-reversal tests, the era ladder, and the concentration among recorded U.S. nationals tie the departures to the sanctions channel directly.

The replacement analysis is descriptive, in that it characterizes who fills vacated seats among matched exit–entry pairs while leaving the causal effect of director loss on replacement type unestimated. The domestic-technocrat result ($+0.0095$, significant at the 10% level) is sensitive to how we define the CCP keyword set, as a strict three-keyword definition weakens it to a coefficient that is not significant at the 10% level, whereas a broad ten-keyword definition flips the sign (-0.006, not significant at the 10% level). The sensitivity follows from the construction of the measure: because the domestic-technocrat indicator is a residual category, the boundary between party-affiliated and party-adjacent careers is inherently ambiguous. Because the replacement-pair matching uses annual data, exact exit–entry pairing is only approximate. The board-size correction nonetheless stands. Using CSMAR's pre-computed board-size measure, the coefficient is null ($+0.007$, not significant at the 10% level), which confirms complete one-for-one replacement and reveals the earlier -0.33-seat shrinkage to have been an artifact of a director-panel count that included supervisory and executive positions. Note that the replacement result also weakens to a coefficient that is not significant at the 10% level when we exclude the 124 early-treated firms ($+0.004$), consistent with those firms driving part of the effect.

The valuation evidence of Section 6 carries the parallel qualification set out there: the education moderation is design-dependent and clears only the ten-percent threshold, so we read the valuation results as consistent with a brain-loss channel rather than as establishing one. The same bundling caveat applies, in that valuation may respond to components of the decoupling shock other than director loss.

The mainland arm shows no robust institutional-ownership response at treated A-share firms across three independent sources, a pattern consistent with a capital-base channel operating only where baseline foreign exposure is substantial; the full mainland ownership nulls appear alongside the Hong Kong capital-base evidence in Appendix A.3. The Hong Kong surname classifier is a noisy proxy, in that it cannot distinguish a Hong Kong permanent resident with a Western surname from a U.S. citizen, and it misclassifies ethnically Chinese directors who hold Western citizenship. Our multi-dialect approach reduces this error without eliminating it, and the independent CSMAR-sourced nationality classifier (70.5% coverage of Hong Kong director-years, with a 25-firm overlap with the surname group) supplies the triangulating cross-check on which we rely throughout. Because the explicit-sanctions sample comprises only 11 firms, all of them

[13] The Persons Rule is scoped to advanced semiconductors, so its escalation should load on chip-sector firms, a sorting no pandemic mechanism produces. We examined this directly by interacting the era-specific estimates with an indicator for the CSRC C39 sector (computers, communications, and other electronic equipment). The interactions carry the predicted signs—the semiconductor differential in the within-board gap is largest in 2023, and the Persons-Rule-era firm-level coefficient is differentially negative for C39 firms—but with only 180 U.S.-nexus directors (689 director-years) at C39 firms none approaches statistical significance, so we report the test as underpowered rather than as evidence.

large SOEs, the strong ownership results may not generalize to smaller or privately owned firms, and the Stock Connect sign reversal between sanctioned and intent-to-treat firms further complicates the ownership interpretation. Several of the cross-market statements we retain lean on the eleven-firm coefficient, and we accordingly report them tentatively.

Several extensions would strengthen the analysis. With the within-board exit-hazard design in Section 3 differencing out firm-level mean reversion directly, and the director-tenure control of Table A8—now estimated on effectively the full panel—confirming that the widening exit gap does not reflect tenure composition, the remaining refinements lie beyond the director panel. Investor-level registry data on Qualified Foreign Institutional Investor (QFII) and Renminbi QFII (RQFII) holdings, beyond the aggregate QFII shares and top-10 approximations we use, would sharpen the capital-base test. A further avenue would ask whether the U.S.-connected directors left around the time their own firm was named for sanction, which would tie the exits more closely still to firm-specific legal exposure if departures followed each listing date. This reading is unavailable in practice. Few of the firms we study were named individually, and fewer still had U.S.-connected directors before the shock, so the firms that could in principle reveal firm-specific timing are too few to support it. The named firms came under the binding investment prohibition at essentially the same moment, in any case, so their listing dates carry little of the spread in timing that such a design would require. That scarcity itself reinforces the broader, exposure-based reading we adopt throughout, which does not turn on resolving any single firm's sanction date.

8 Conclusion

One mechanism runs through the evidence. Heightened personal legal exposure under the U.S. Persons Rule raised the private cost of board service for U.S.-connected directors, and boards adjusted by shedding them—selectively under intent-to-treat exposure, where the elevated exit is confined to U.S.-connected directors, and indiscriminately under explicit sanction, where foreign directors of all origins decline and international capital reallocates, though this indiscriminate pole rests on the eleven explicitly sanctioned firms and we read it cautiously. Because the identification rests on a within-board comparison, in which each U.S.-connected director is measured only against the non-U.S. colleagues serving on the same board in the same year, the exit survives the firm-level confounders—mean reversion, term cycles, and a market-wide de-internationalization—that a share-based difference-in-differences could not rule out. The same mechanism reproduces across the mainland A-share and Hong Kong markets, which differ in regulation, ownership, and the depth of the labor pool that absorbs the shock, and the replacement pattern moves with that depth, as mainland boards recruit domestic professionals while Hong Kong boards recruit non-U.S. internationals.

These findings identify board composition as one internal channel through which geopolitical fragmentation reaches the firm, distinct from the trade and financial channels prior work emphasizes and operating through the market for director human capital. The policy reading is pointed. A measure aimed at restricting U.S. capital flows reshaped the human capital of Chinese boards instead, removing internationally credentialed independent monitors of the kind associated with stronger minority-shareholder protection. The near-term consequences we can measure are bounded, since one-for-one domestic substitution left the monitoring outcomes we observe undisturbed; whether harder-to-observe channels that prices do not capture eventually register, and whether the costs compound over a longer horizon, are questions the enforcement and balance-sheet data now being assembled will be needed to answer. The one margin on which a cost registered at all was valuation: Tobin's Q declines at exposed privately owned firms while the state sector shows no detectable movement, and the attribution of that decline to the departing directors' human capital is suggestive, design-dependent and clearing only the ten-percent threshold. The cost was detectable at private firms rather than at the state-linked entities the sanctions targeted, because access to internationally credentialed directors is a market relationship that no state guarantee replaces.

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A Appendix

A.1 Measurement Validation

Our empirical exercise requires an attribute that standard corporate-governance data do not report directly, namely whether an individual director holds a meaningful connection to the United States. We construct this attribute through a staged measurement procedure whose primary layer rests on auditable structured data and is supplemented, for the cases that structured fields cannot resolve, by a classification layer.

Our primary measure classifies a director as U.S.-nexus whenever any of three conditions holds in CSMAR structured records: (i) U.S. nationality or permanent residency, (ii) at least five years of U.S. work experience, or (iii) a U.S. professional license. We apply this definition to the resume, education, professional-title, nationality, and licensure fields of the CSMAR Corporate Governance module. The U.S.-professional-license prong is matched with guarded patterns that exclude non-U.S. credentials: the token "CPA" must stand free of adjacent letters, so that it does not fire inside compound abbreviations such as CICPA, FCPA, CCPA, or ICPA; the Chinese licensure phrase for a U.S.-registered credential is matched only where it does not continue into the phrase denoting the U.S. Certified Management Accountant, which is not a CPA license; and explicit U.S.-license forms (USCPA, AICPA, and the Chinese rendering of "U.S. CPA") are retained. We manually audited every director whose U.S.-license flag changed under this refinement, confirming that the 215 directors it de-flagged held non-U.S. credentials and that the 17 rescue-matched directors are genuinely U.S.-credentialed. Because this structured measure is fully auditable and does not depend on any model-based inference, we treat it as the regressor underlying all of our headline estimates.

For directors whose U.S. connection we cannot verify from structured fields, we supplement the structured measure with biographical text retrieved from public web sources (i.e., Baidu search snippets), which a large-language model then classifies into a nationality / U.S.-nexus label. This procedure yields the market-specific classified files that we use in our robustness analysis. We treat the classification layer as supplemental, since it defines the expanded treatment definitions (Specifications 2 and 3 in Appendix A.4) while leaving the primary measure unchanged. As Appendix A.4 documents, the headline exit estimate attenuates only mildly once we add these classified directors (from -0.021 to -0.018), which confirms that the result survives the addition of model-based labels. We therefore present the auditable three-prong construct as our primary measurement approach and the classification layer as a robustness extension; Appendix A.2 sets out the retrieval and classification procedure in full, together with evidence that vendor coverage is uncorrelated with treatment (Table A1).[14]

In the Hong Kong market, where structured nationality coverage is thinner, we infer director

[14] Because the classification layer enters only the expanded treatment definitions (Specifications 2 and 3) and leaves the headline estimates essentially unchanged, we treat its full reproducibility and accuracy validation—fixed-temperature reruns and precision and recall against a hand-labeled audit sample—as supplemental, to be reported should a referee ask that the classifier be foregrounded.

origin from a multi-dialect Chinese surname classifier (i.e., a classifier spanning Mandarin, Cantonese, Hokkien, and Hakka) that flags non-Chinese names. We validate this surname measure against an independent CSMAR-sourced nationality classifier, which covers 70.5% of Hong Kong director-years. The two instruments appear to capture largely distinct director populations, since the surname-based and nationality-based treatment groups overlap on only 25 firms and neither treatment predicts the other's dependent variable (cross-DV checks, not significant at the 10% level). We read this pattern as triangulation, because the classifiers measure distinct populations and thereby serve as mutual cross-checks. The surname classifier is admittedly noisy, since it cannot distinguish a Hong Kong permanent resident bearing a Western surname from a U.S. citizen, and it misclassifies ethnically Chinese directors who hold Western citizenship. We therefore treat the surname measure as a broad indicator of Western-aligned directors and the nationality measure as the narrower, U.S.-specific instrument; in Section 5 we exploit the gap between the two to separate the U.S.-specific channel from broader de-internationalization.

A.2 Director-Nationality Classification

This appendix documents, step by step, how we assign a nationality to every director in the sample and how the U.S.-nexus flag underlying the treatment is built from that assignment. The procedure has two layers. The first applies deterministic rules to CSMAR structured records and defines the primary measure behind all headline estimates. The second covers directors for whom CSMAR records no nationality: for these individuals we retrieve biographical snippets from Chinese web search and classify them with a large-language model. A recorded CSMAR value always takes precedence over a model label, so the second layer only fills gaps left by the first.

A.2.1 Structured Classification from CSMAR Records

Director characteristics come from two CSMAR files joined at the person level. The annual Corporate Governance roster supplies resume text, position, education, and the overseas-background indicator; it contains no nationality field. A second, report-date director-profile file records director attributes at each disclosure date and is the sole source of recorded nationality; it also supplies the longest available resume for each person, which we use to backfill short roster resumes. Nationality appears in the profile file as a raw Chinese-language country string, and we take each person's most recent non-missing value. After merging this person-level lookup onto the annual roster, nationality is available for 66.7% of director-years in the full 2018–2024 roster, while 44,396 unique directors carry no recorded nationality at all. For the nationality-based composition outcomes we map the recorded strings into four classes—Chinese, U.S., Hong Kong (with Macau), and other foreign, the last covering roughly thirty country strings, Taiwan among them—and the non-U.S.-foreign share used in the placebo and in Figure 4 counts the other-foreign and Hong Kong classes together.

The U.S.-nexus flag is the union of the three prongs introduced in Section 3.1, evaluated on these fields. Prong 1 flags a director whose recorded nationality is the United States; it contributes

443 individuals. Prong 2 requires three conditions jointly: the overseas-background indicator records overseas work experience; the resume mentions a U.S. location or institution, drawn from a keyword list spanning major U.S. cities, Wall Street financial institutions, U.S. regulators and exchanges, elite U.S. universities, and large U.S. technology and professional-services firms; and the resume ties a U.S. affiliation to an explicit year span of at least five years. Prong 3 flags any director whose resume matches the guarded U.S. professional-license patterns whose audit Appendix A.1 describes, and it applies to all directors whether or not the overseas-background indicator is set. A director flagged in any year is coded U.S.-nexus in every year, since the underlying attribute is durable. Net of prong 1, prongs 2 and 3 together contribute a further 849 individuals, so the three prongs flag 1,292 unique directors (8,180 director-year observations), and aggregation to the firm level—the flagged share of the board, and a treatment indicator for at least one flagged director in the 2019–2020 baseline—yields the 1,040 treated firms of the primary sample.

A.2.2 Search-Based Classification of Directors Without Recorded Nationality

Before describing the classification, we document that the incompleteness of the vendor's nationality field cannot drive our results. Table A1 collects the evidence. The input on which the primary three-prong measure rests—resume text—is available for 99.2% of director-years and is uncorrelated with treatment in levels and in trends (row 1: $p = 0.34$ and $p = 0.94$). Coverage of the nationality field itself is disclosure-driven, since biographies state nationality where it is informative, and internationally connected directors are accordingly better covered: within their own boards, U.S.-nexus directors identified from resume text alone are 4.0 percentage points more likely than their boardmates to carry a recorded nationality, a differential that does not move at the shock (row 2: $p = 0.40$). Missingness therefore concentrates among domestic directors—94.7% of the unrecorded pool classifies as PRC nationals below—while prong 1 can only ever fire inside the recorded pool and prongs 2 and 3 are evaluable nearly everywhere. The same pattern holds in Hong Kong, where the surname instrument is computed from the director's name and is defined for every director by construction, and where the resolution rate of the search-based cross-check is unrelated to the Western-surname treatment in levels and trends (row 3: $p = 0.32$ and $p = 0.83$). Coverage, then, cannot drive the results; in any case, the search-based classification below includes the uncovered directors, and the headline exit estimate moves only from -0.021 to -0.018 when they are added (Table A6).

Table A1: Vendor Coverage, Search Resolution, and Treatment

	(1) Level vs. treatment	(2) Treated $\times$ Post
Mainland: board share with resume text	$+0.0007$	$+0.0001$
	(0.340)	(0.942)
Mainland: recorded nationality, U.S.-nexus director (director level, within firm $\times$ year)	$+0.0402^{***}$	$+0.0132$
	(0.008)	(0.400)
Hong Kong: board share with resolved nationality class	$+0.0180$	$+0.0021$
	(0.323)	(0.828)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. Column (1) reports the association between each coverage measure and treatment in levels (firm-year rows: year fixed effects and firm controls; director-level row: the pooled within-firm$\times$year gap between U.S.-nexus directors and their boardmates). Column (2) reports the post-2021 differential (firm-year rows: Treated $\times$ Post with firm and year fixed effects; director-level row: the U.S.-nexus $\times$ Post interaction). Prong 1 is excluded from the director-level row because its link to recorded nationality is definitional. Estimation samples: 19,602 (19,593) mainland firm-years in column (1) (column (2)), 374,614 mainland director-years, and 16,370 (16,312) Hong Kong firm-years; base coverage rates are 99.2% (resume text), 66.7% (recorded nationality), and 70.9% (Hong Kong resolved class). The recorded-nationality share itself is balanced in levels ($+0.018$, $p = 0.12$; yearly interactions in 2018–19 within ± 0.002) and declines by 0.9 percentage points at treated firms after 2021 ($p = 0.011$); this differential reflects the board reshuffle itself, operating through the departure of foreign directors whose nationality is recorded by construction (-0.54 pp, $p < 0.001$) and through post-shock appointees newer to the disclosure ecosystem (entrant volumes are identical at 13.5% of the roster), and it bounds any bias in the nationality-based composition outcomes at roughly 0.03 percentage points. In Hong Kong the U.S.-nationality and sanctioned treatments show higher resolution in levels ($+0.043$, $+0.059$), which is mechanical—the former defines treatment from resolved classes and the latter comprises large firms with dense web coverage—with no differential trend ($p = 0.26$, $p = 0.67$).

For the 44,396 directors with no recorded nationality we retrieved biographical evidence from Chinese web search in March 2026, issuing queries through a commercial search interface (SearchAPI) against the Baidu engine. The baseline query pairs the director's name with the Chinese keywords for "director," "nationality," and "resume." When a first pass returned an unknown label or low confidence, the pipeline re-queried under two progressively broader templates—the name with resume keywords only, then the bare name—and kept the most informative classification across passes. Each query retains up to twelve organic results (title, link, and snippet). Snippets from Sina Finance's listed-company profile pages are placed first, up to five, because those pages reproduce standardized director biographies from disclosure filings; the remaining slots are filled from the other organic results, which in practice include Baidu Baike, Baidu's business-registry service, cninfo, and Eastmoney.

A large-language model (DeepSeek's "deepseek-chat," called with JSON-constrained output) then classifies each director from the retained snippets into one of five labels: Chinese, U.S., Hong Kong, other foreign, or unknown. The fixed instruction imposes an evidence hierarchy: explicit nationality or citizenship statements rank first, then place of birth, then education (in particular the undergraduate institution and its country), then long-run career location. It also builds in guardrails for this setting. The model may not infer Chinese nationality from a Chinese name or from service on a Chinese board, because many overseas Chinese hold foreign passports; dual nationals are classified by their primary nationality; directors described as foreign without a stated country go to the other-foreign class; and the output must contain the label, a three-level confidence grade, the specific country where applicable, and a one-sentence citation of the decisive evidence. A director whose searches return nothing defaults to unknown without a model call, and malformed model output is likewise coerced to unknown. Of the 44,396 directors processed, the model classified 42,032 (94.7%) as Chinese nationals, 347 (0.8%) as U.S., 461 as Hong Kong, and 851 as other foreign nationals, leaving 705 (1.6%) unresolved.

A.2.3 Scope, the Hong Kong Arm, and Limitations

These person-level labels enter the analysis at three points, in each case subordinate to the structured layer. First, they complete the nationality classes behind the composition outcomes and the non-U.S.-foreign placebo; wherever CSMAR records a nationality the recorded value prevails, and unresolved directors are excluded from the foreign counts. Second, the model-classified U.S. nationals define the first expanded treatment definition, Specification 2 (+57 firms; Table A6). Third, the profile file contains 23,704 directors who never appear in the annual roster and for whom CSMAR supplies neither resume text nor the overseas-background indicator, so prongs 2 and 3 cannot be evaluated for them from structured fields. A second, search-free pass therefore re-scored the archived snippets of these directors against the three-prong definition, using the same model at a sampling temperature of 0.1 on the 2,166 candidates that survived a U.S.-keyword screen; it identified 511 U.S.-nexus directors (269 by nationality, 221 by a U.S. work history of at least five years, and 21 by a U.S. license), and these define the Specification 3 expansion (+138 firms). The primary treatment uses the structured layer alone.

The Hong Kong arm follows the same two-step architecture with market-appropriate sources. Queries pair the director's name with the company name and English-language keywords and run against the Google engine; snippets from the exchange's disclosure archive (HKEXnews), Webb-site, Bloomberg, and Reuters are prioritized; and the classification prompt additionally receives the multi-dialect surname romanization of Appendix A.1 as an explicitly weak prior. Of the 42,545 Hong Kong directors processed, roughly 40% remain unresolved, a reflection of the thinner biographical coverage of Hong Kong directors in public web sources. The unresolved share is balanced across treatment groups: firm-year resolution rates are unrelated to the Western-surname treatment in levels and in trends (Table A1), and the surname instrument itself, computed from the director's name, is defined for every director by construction. This noise is why Section 5 treats the surname classifier as the broad instrument and the nationality-based classifier as the narrower, U.S.-specific cross-check.

Two limitations of the model layer deserve note. Queries are keyed on the director's name, so common Chinese names carry a residual risk of conflating namesakes, and the evidence consists of search snippets rather than full source documents, so the year spans required by the five-year work rule are sometimes unstated and push the classifier toward a medium confidence grade. The design responds to both risks conservatively: unresolved and low-confidence cases stay in the unknown class and drop out of all foreign counts, recorded CSMAR values override model labels wherever the two coexist, and the layer is quarantined from the primary treatment, whose headline exit estimate moves only from -0.021 to -0.018 when the model-classified directors are added (Table A6).

A.3 Capital-Base Response and the Compliance-Chill Threshold

This appendix collects the cross-market capital-base evidence and the full selective-versus-indiscriminate synthesis demoted from the main text, together with the corroborating departure-return evidence that triangulates Section 6. We present this material here because it is auxiliary to the within-board exit design that carries the main identification claim.

A.3.1 International Ownership: Explicitly Sanctioned Firms

Table A2: International-Ownership Decline: Explicitly Sanctioned Hong Kong Firms

	(1) International	(2) International	(3) Mainland	(4) Stock Connect
DV: Ownership share (%)				
Sanctioned × Post	-11.05^{***}	-10.36^{***}	$+3.04^{**}$	$+11.96^{***}$
	(<0.001)	(<0.001)	(0.020)	(0.004)
Controls	No	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Observations	79,495	71,510	71,509	20,120
Firms	2,398	2,307	2,307	739
R^2 (within)	-0.0003	0.0041	0.0005	0.1038

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table uses the Hong Kong quarterly ownership panel for explicitly sanctioned firms and reports ownership-share outcomes with firm and quarter fixed effects.

Table A2 tests whether explicitly sanctioned Hong Kong-listed firms experienced a decline in international institutional ownership. Treatment is binary, comprising the 11 firms designated on U.S. sanctions lists. The quarterly panel comprises 89,621 firm-quarter observations (2,398 firms) in the raw panel, with ownership breakdowns sourced from Wind Terminal, while the estimation samples are smaller after controls and coverage restrictions (Table A2). Sanctioned firms experienced a 10.4 pp decline in international ownership (significant at the 1% level), accompanied by offsetting increases in mainland Chinese ownership ($+3.0$ pp, significant at the 5% level) and

Stock Connect holdings ($+12.0$ pp, significant at the 1% level). The reallocation appears to operate through the northbound/southbound Connect mechanism, with direct mainland purchases playing little role. In the broader intent-to-treat sample, by contrast, ownership effects are uniformly null for international and Chinese ownership (Table A3; all not significant at the 10% level), leaving only a negative Stock Connect effect. Measurable financial-capital reallocation thus appears only among firms subject to legally binding investment prohibitions, although the intent-to-treat nulls partly reflect lower baseline foreign exposure and therefore provide limited evidence on investor non-response.

A.3.2 International Ownership: Intent-to-Treat

Table A3 extends the ownership analysis to the broader intent-to-treat sample, using the Western-surname and foreign-dialect treatment definitions. The results are uniformly null for international and Chinese ownership, since none of the four coefficients approaches significance (not significant at the 10% level). The only significant result is a negative Stock Connect effect (-2.47, significant at the 1% level for Western treatment; -3.12, significant at the 1% level for foreign-dialect). This sign reversal relative to the explicitly sanctioned firms (Table A2) likely reflects compositional differences. Explicitly sanctioned firms are predominantly large SOEs that attract Stock Connect inflows as mainland investors substitute for departing international holders, whereas the broader intent-to-treat sample includes smaller firms in which foreign-director presence signals international orientation without triggering forced ownership reallocation. Ownership concentration (top-10 shareholder share) is null across all specifications (not significant at the 10% level). These nulls constitute the financial-capital half of the threshold result: measurable capital reallocation appears to require the binding prohibition of explicit sanctions, while mere geopolitical exposure yields null ownership effects and limited evidence, because baseline foreign exposure is low.

A.3.3 Mainland Ownership: No Measurable Response

The mainland arm finds no robust institutional-ownership shift at treated A-share firms. Three data sources point the same way: a top-10 shareholder origin decomposition, aggregate institutional ownership (CSMAR), for which the estimate is small and at most marginally negative, and QFII-specific foreign ownership (-0.02pp, $p = 0.50$). The QFII data strongly support the structural-exposure interpretation, since sanctioned A-share firms held only 0.06% QFII pre-shock, leaving little QFII capital to lose. One qualification tempers this null: the top-10 analysis shows a small increase in Hong Kong nominee/HKSCC clearing-account ownership ($+0.16$, significant at the 5% level), likely reflecting Stock Connect reallocation, though it does not survive as a robust aggregate finding. The mainland null is consistent with a capital-base substitution channel that operates only where baseline foreign exposure is substantial, which is why a measurable effect appears only among explicitly sanctioned Hong Kong-listed firms, where baseline international ownership stood at approximately 26%.

Table A3: International Ownership: Intent-to-Treat (Hong Kong Quarterly Panel)

	(1)	(2)	(3)	(4)	(5)	(6)
	DV: International ownership (%)		DV: Mainland ownership (%)		DV: Stock Connect ownership (%)	
	Western	Foreign	Western	Foreign	Western	Foreign
Treated × Post	$+2.42$	$+0.97$	$+1.36$	-0.51	-2.47^{***}	-3.12^{***}
	(0.226)	(0.215)	(0.227)	(0.558)	(<0.001)	(<0.001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm
Observations	71,510	71,510	71,509	71,509	20,120	20,120
Firms	2,307	2,307	2,307	2,307	739	739
R^2 (within)	0.0078	0.0113	0.0005	0.0004	0.0601	0.0170

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports broader intent-to-treat ownership estimates for Western-surname and foreign-dialect treatment definitions in the Hong Kong quarterly panel.

A.3.4 The Compliance-Chill Threshold

The cross-market evidence is consistent with a single mechanism operating at different intensities of legal exposure. Under compliance chill (i.e., intent-to-treat exposure with no binding prohibition), the response is selective: the elevated exit is confined to U.S.-connected directors, whereas non-U.S. foreign directors show no detectable change on the mainland and rise in Hong Kong. Under legal compulsion through explicit sanctions, the response becomes indiscriminate, as foreign directors of all origins decline and international capital reallocates. Table A4 lays the three regimes side by side.

Table A4: Selective Compliance Chill versus Indiscriminate Sanction: Cross-Market Board and Ownership Response

	Mainland (ITT)	HK (ITT)	HK (Sanctioned)
U.S./Western director exit	-0.021^{***}	U.S.-nat. -0.013^{**}; surname -0.025^{***}	-0.014 (0.309)
Non-U.S. foreign response	$+0.0005$ (0.651)	$+0.010^{**}$	-0.029^{*}
Replacement type	Domestic technocrat	Non-U.S. international	Not directly measured; foreign share declines
Intl. ownership response	Null; QFII baseline $\approx$0.27%	Null; Stock Connect -2.47^{***}	Intl -10.4^{***}; CN $+3.0^{**}$; Connect $+12.0^{***}$
Selectivity	Selective	Selective (substitution)	Indiscriminate

Note: Cell entries are difference-in-differences coefficients on director-share or ownership-share outcomes; parenthetical notes give significance or classifier. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Treated firms: mainland 1,040; Hong Kong ITT 137 (U.S.-nationality); Hong Kong sanctioned 11. The sanctioned U.S.-nationality cell is imprecise ($n = 11$); the indiscriminate reading rests on the significant non-U.S. (-0.029^{*}) and international-ownership (-10.4pp) cells rather than that point estimate. Sources: Tables 3, 8, 9, 12, A2, 11, and A3.

The contrast indicates that human capital and financial capital respond at different thresholds of geopolitical pressure: director exit appears across the full intent-to-treat sample, whereas measurable ownership reallocation appears only where transacting in a specific security became legally prohibited. We read this as suggestive evidence, since the ownership nulls partly reflect negligible baseline foreign exposure—A-share sanctioned firms held only about 0.06% QFII pre-shock, which limits what they can reveal about investor response—and "compliance chill" is our interpretation inferred from the director pattern.

A.3.5 The Market's Reaction at Departure

This appendix reports in full the consequence test that Section 6 alludes to but does not tabulate in the main text: the market's reaction to U.S.-connected director departures. The test returns a bounded null consistent with the crash-risk, pay-performance, and payout results reported there.

We first ask how the market priced the departures themselves. The events are U.S.-connected director departures matched to the director panel, with informative departures separated from scheduled term-expiries. Abnormal returns come from a one-factor market model estimated on log returns over the $[-250, -30]$ window of at least 120 trading days, applying a trade-to-trade correction for the thin trading common among Chinese stocks.[15] Cumulative abnormal returns (CARs) are then aggregated over symmetric windows after collapsing multiple departures at the same firm and date. Inference draws on a battery of tests: the Patell and Boehmer-Musumeci-Poulsen standardized-residual tests, the generalized sign test benchmarked on the estimation window, the Wilcoxon signed-rank test, and the cross-sectional Student-t, supplemented by Kolari-Pynnonen cross-correlation-adjusted and GRANK-t variants. Were the loss of a U.S.-connected monitor read as value-destroying, one would expect a negative and significant cumulative abnormal return at the departure date; instead the cumulative abnormal returns are economically small and statistically indistinguishable from zero across this battery (Table A5), so the market registered no robust reaction to the loss of U.S.-connected directors. Panel A reports cumulative abnormal returns by departure group across event windows under the primary market model, where the magnitudes are small and the scattered significance in the scheduled and pooled groups tracks term-expiry calendar clustering with no U.S.-specific reaction. Panel B shows that the only positive flicker, the informative independent departures over $[-2, 2]$, is model-sensitive. The cumulative abnormal return is stable in magnitude across market benchmarks and model specifications, yet it attains significance only under the three-factor model, and even there only at conventional-to-marginal levels; under the primary thin-trading market model it fails every test in the battery.

[15] The thin-trading correction follows Maynes and Rumsey (1993).

Table A5: Director-Departure Cumulative Abnormal Returns

Panel A. CAARs by departure group and event window (market model, main-board A-share benchmark)

	$[0,0]$	$[-1,1]$	$[-2,2]$	$[-5,5]$	$[-10,10]$
U.S.-nexus informative, post (N=270)	-0.01	$+0.12$	$+0.34$	$+0.03$	$+0.11$
of which independent (N=86)	$+0.21$	$+0.45$	$+1.09$	-0.27	$+1.10$
U.S.-nexus all departures, post (N=779)	-0.09	-0.03	$+0.20$	$+0.23$	-0.10
U.S.-nexus scheduled, post (N=510)	-0.14^{*}	-0.12	$+0.11$	$+0.33$	-0.27
U.S.-nexus informative, pre (N=156)	$+0.19$	$+0.21$	-0.33	-1.06	-1.13

Panel B. Informative independent departures (post; $N = 86$): the $[-2,2]$ reaction across models and benchmarks

	CAAR%	Student-t	Patell	BMP	Sign	Wilcoxon
Market model, main-board A-shares	$+1.09$	1.28	1.18	0.77	0.91	0.82
Market model, broad A-shares	$+1.11$	1.32	1.17	0.77	0.99	0.80
Market model, ST/halt-excluded	$+0.87$	0.99	0.94	0.61	0.79	0.58
Three-factor, main-board A-shares	$+1.40$	1.78*	2.36**	1.79*	1.64	0.78
Three-factor, broad A-shares	$+1.47$	1.87*	2.37**	1.76*	1.38	0.89

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Cumulative abnormal returns (CAARs) are in percent, averaged across departure events after collapsing multiple departures at the same firm and date; event day 0 is the director's recorded departure date, as CSMAR reports no separate announcement date. Abnormal returns are from a one-factor market model (Panel A and the market-model rows of Panel B) or a Fama-French three-factor model (the three-factor rows), estimated on log returns over $[-250, -30]$ (minimum 120 trading days) with a trade-to-trade thin-trading correction (Section 6). "Main-board A-shares" is the Shanghai-plus-Shenzhen A-share market excluding the ChiNext and STAR boards, and "broad A-shares" adds those boards. In Panel A, significance markers are from the Patell test, and the "of which independent" row is the independent-director subset of the informative-departure group. Panel B reports test *statistics*: Student-t is the cross-sectional t; Patell and BMP are the standardized-residual tests; Sign is the generalized sign test benchmarked on the estimation window; and Wilcoxon is the signed-rank statistic. For the primary market-model focal cell the Kolari-Pynnonen-adjusted Patell statistic (0.34) and the GRANK-t statistic (1.15) are likewise insignificant, and a cross-sectional regression of the $[-2,2]$ abnormal return on the U.S.-nexus-by-post interaction yields $+0.28$ percentage points ($p = 0.68$). The benchmark adding only ChiNext (excluding STAR) lies between the two reported benchmarks and yields the same pattern.

The remaining proxies that bear on channels less visible to prices—the incidence of enforcement actions and financial restatements, and abnormal other-receivables as a tunneling measure—require enforcement and balance-sheet data still being assembled, and we leave them as future evidence. We therefore treat them as the next natural step for evaluating the consequence channel beyond return-based measures.

A.4 Robustness and Pre-Trend Diagnostics

A.4.1 Treatment-Definition Robustness

Table A6: Treatment-Definition Robustness: Main Results and Placebo Tests

	Spec 1 N_T=1,040	Spec 2 N_T=1,097	Spec 3 N_T=1,235
Panel A: Main Results			
Director exit (share)	-0.0209^{***}	-0.0190^{***}	-0.0183^{***}
	($<$0.001)	($<$0.001)	($<$0.001)
Director exit (count)	-0.3860^{***}	-0.3487^{***}	-0.3443^{***}
	($<$0.001)	($<$0.001)	($<$0.001)
Replacement (domestic technocrat)	$+0.0095^{*}$	$+0.0095^{**}$	$+0.0092^{*}$
	(0.058)	(0.046)	(0.051)
Panel B: Placebo Tests			
Placebo T1 (same firms)	$+0.0005$	$+0.0008$	$+0.0009$
	(0.651)	(0.459)	(0.386)
Placebo T2 (unexposed)	-0.0158^{***}	-0.0162^{***}	-0.0167^{***}
	($<$0.001)	($<$0.001)	($<$0.001)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. Specification 1 uses the CSMAR three-prong heuristic; Specification 2 adds Baidu-scraper-classified U.S. nationals; Specification 3 adds scraper work experience and licenses. Placebo T1 uses non-U.S. foreign share at treated firms, and T2 uses non-U.S.-foreign-treated firms with no U.S. exposure. All specifications include firm and year fixed effects and firm-clustered standard errors; the exit and replacement rows add the controls of Section 3, while the placebo rows are estimated without the firm controls.

Table A6 reports our results under three treatment definitions. The primary specification (Specification 1) relies on CSMAR structured data to classify U.S.-nexus directors via the three-prong heuristic, identifying 1,040 treated firms. The expanded definitions add directors recovered from Baidu search snippets, so that Specification 2 incorporates scraper-classified U.S. nationals (+57 firms), while Specification 3 further adds directors with U.S. work experience or U.S. licenses (+138 firms).

We adopt Specification 1 as our primary specification because the expanded definitions introduce measurement noise. The scraper-identified directors differ systematically from the CSMAR-identified population: they are predominantly non-independent (75–78% vs. 52%), they exhibit weaker post-shock exit rates (-25.5% vs. -41.9% at treated firms over 2020–2024), and 73–74% have zero observations in the analysis window, having served before the sanctions period. These characteristics are consistent with the scraper capturing less prominent directors whose U.S. connections are harder to verify from structured data.

Our results remain robust across all three definitions. Panel A shows that the exit estimate attenuates monotonically from -0.021^{***} (Specification 1) to -0.018^{***} (Specification 3), consistent with adding firms whose U.S.-nexus directors exit at lower rates. The replacement result is essentially unchanged across definitions ($+0.010$*, Specification 1, to $+0.009$*, Specification 3), while Panel B confirms that the placebo tests are invariant to the treatment definition.

A.4.2 Cross-Specification Robustness Summary

Table A7 consolidates the key coefficient estimates across our retained hypotheses and extensions. Panel A reports the main results, whereas Panel B reports the extensions, including the corrected board-size result ($+0.007$, null) and the independent-director ratio (-0.098, null), together with the SOE executive-pay decline (-0.045, significant at the 10% level). Panel C reports industry $\times$ year fixed-effect specifications, under which the exit result is virtually unchanged (-0.021***). Panel D reports the most conservative clustering. Because the baseline clusters on roughly 2,911 firms, asymptotic inference is already reliable at the firm level, so we present the more demanding clustering schemes as reassuring sensitivity checks. The exit result remains highly stable under two-way firm and year clustering—itself a conservative sensitivity check, since the year dimension carries only six clusters—where the standard error increases roughly fourfold yet remains significant at the 1% level, and the replacement result survives. As a further check at the dimensions where cluster counts are small enough to threaten the asymptotic approximation, the wild cluster bootstrap confirms the exit result (significant at the 1% level) under both industry (18 clusters) and province (36 clusters) bootstraps.

Table A7: Cross-Specification Robustness Summary

	DV	$\hat{\beta}$	p	N	Sample
Panel A: Main Results					
H1	U.S.-nexus director share	-0.0209^{***}	<0.001	16,987	2019–2024
H2	Domestic technocrat	$+0.0095^{*}$	0.058	310,645	2019–2024
Panel B: Extensions					
Ext 1a (dose)	U.S.-nexus director share	-0.3860^{***}	<0.001	16,987	2019–2024
Ext 1b	Euphemistic resignation	-0.0763^{***}	0.004	75,674	2019–2024
Ext 1c (Independent)	U.S.-nexus share (independent)	-0.0523^{***}	<0.001	16,987	2019–2024
Ext 2a (Technical-background index)	Technical-background index	$+0.0069^{***}$	0.004	310,645	2019–2024
Ext 2b (Board size)	Board size	$+0.0074$	0.856	16,926	2019–2024
Ext 2b (Independent ratio)	Independent-director ratio	-0.0984	0.566	16,925	2019–2024
Ext 2c (POE indep)	Log(indep. director pay)	$+0.1109$	0.120	5,334	2019–2024
Ext 2c (TOP3, SOE)	Log(top-3 exec. pay)	-0.0454^{*}	0.069	5,454	2019–2024
Panel C: Industry $\times$ Year FE Robustness					
H1 (Ind$\times$Yr)	U.S.-nexus director share	-0.0207^{***}	<0.001	16,987	2019–2024
Panel D: Alternative Clustering (Most Conservative)					
H1 (Firm+Year)	U.S.-nexus director share	-0.0209^{***}	<0.001	16,987	2019–2024
H2 (Firm)	Domestic technocrat	$+0.0095^{*}$	0.058	310,645	2019–2024

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. All specifications include firm and year fixed effects unless noted; the replacement analysis uses replacement-pair data with the treated-U.S.-exit indicator as the treatment variable.

A.4.3 Director-Tenure Robustness

Table A8: Director Tenure and the Within-Board Exit Gap

	(1) Baseline	(2) + Tenure control
U.S.-nexus × 2018	-0.0324^{***}	-0.0221^{**}
	(0.001)	(0.024)
U.S.-nexus × 2019	$+0.0195^{*}$	$+0.0236^{**}$
	(0.081)	(0.028)
U.S.-nexus × 2021	$+0.0265^{**}$	$+0.0220^{**}$
	(0.022)	(0.044)
U.S.-nexus × 2022	$+0.0455^{***}$	$+0.0398^{***}$
	(<0.001)	(<0.001)
U.S.-nexus × 2023	$+0.0434^{***}$	$+0.0355^{***}$
	(0.001)	(0.003)
Firm × Year FE	Yes	Yes
Director-tenure control	No	Yes
Director-years	313,095	313,095
Firms (clusters)	2,911	2,911

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values from standard errors clustered by firm. The dependent variable equals one if a director leaves the board before the following year. Each coefficient is the within-board difference, between U.S.-connected directors and their non-U.S. colleagues on the same board in the same year, by calendar year relative to the 2020 base. Both columns include firm × year fixed effects, so identification rests only on differences among directors serving on the same board in the same year. Column 2 adds the director's board tenure, in years served, and its square. The two columns are estimated on the director-years with available appointment dates, which cover 97.9% of director-years across all 2,911 sample firms.

A further concern is that U.S.-connected directors may leave at higher rates because they hold younger directorships, which turn over more often for reasons unrelated to the sanctions. Because U.S.-nexus directors do serve shorter tenures than their board colleagues—a pre-2021 average near two-and-a-half years against three—we remove this channel directly. Table A8 re-estimates the within-board event study with a control for director tenure and its square. The post-shock widening of the within-board exit gap survives, the 2022 and 2023 coefficients moving from $+0.046^{***}$ and $+0.043^{***}$ to $+0.040^{***}$ and $+0.036^{***}$, so the rising gap does not appear to reflect tenure composition. The control leaves a 2019 pre-shock coefficient of $+0.024^{**}$, a within-board divergence we attribute to the early-treated firms discussed in Section 7, and the negative 2018 coefficient in both columns mirrors the same pattern; the within-board design does not in any case rest on pre-trend parallelism. Appointment dates now cover 97.9% of director-years across all 2,911 sample firms, so the control no longer entails a subsample restriction.

A.4.4 Departure Timing, the Reopening, and Cohort Depletion

Table A9 collects the departure-timing evidence used in Sections 4.1 and 7, and Table A10 reports the cohort view. Dated by exact departure date, the within-board U.S.-nexus gap is null before the first investment prohibition took effect, opens during the investment-sanctions era, and peaks under the Persons Rule, with the recorded-U.S.-national gap largest in the final era; the no-reversal panel shows the 2023 gap surviving the border reopening at its 2022 level, and within the Persons-Rule era the gap is, if anything, larger in the reopened months. In the frozen 2019 cohort, U.S.-connected directors depart at significantly higher rates in every era, and the per-year within-firm gap rises from 2.3 percentage points in 2020 to 2.6 in the investment-sanctions era and 3.7 in the Persons-Rule year, although the era-on-era escalation is not itself statistically significant; the depletion visible in the cumulative row is what softens the late-era contrasts in the conditional hazard.

Table A9: Departure Timing by Legal Era and No-Reversal Tests

	(1) Full sample	(2) Recorded U.S. nationals
Panel A: Within-board departure gaps by legal era (exact departure dates)		
Pre-sanctions era	$+0.0072$	$+0.0275^{**}$
	(0.216)	(0.016)
Investment-sanctions era	$+0.0209^{***}$	$+0.0205^{**}$
	(<0.001)	(0.036)
Persons-Rule era	$+0.0275^{***}$	$+0.0411^{***}$
	(<0.001)	(0.007)
Panel B: No-reversal tests and the reopening cut		
Gap in 2023 (post-reopening year) > 0	$+0.0434^{***}$	$+0.0637^{**}$
	(<0.001)	(0.013)
Gap in 2023 $-$ gap in 2022	-0.0023	$+0.0220$
	(0.895)	(0.557)
Gap in 2023 $-$ mean gap, 2020–21	$+0.0167$	$+0.0100$
	(0.136)	(0.385)
Persons-Rule-era gap, zero-COVID months	$+0.0063^{*}$	$+0.0068$
	(0.061)	(0.333)
Persons-Rule-era gap, reopened months	$+0.0211^{***}$	$+0.0346^{**}$
	(<0.001)	(0.011)
Firm $\times$ Year FE	Yes	Yes

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm, and every estimate carries firm $\times$ year fixed effects on the director-level hazard panel. Column 1 compares U.S.-nexus directors with their non-U.S. boardmates; column 2 compares directors with recorded U.S. nationality against never-U.S. boardmates, dropping directors flagged only by the resume or licensure prongs. In Panel A each exit is dated by its recorded departure date (available for 96.9% of exits; undated exits are treated as non-era departures) and assigned to the era in which it falls: the pre-sanctions era runs through January 10, 2021; the investment-sanctions era from January 11, 2021 to October 6, 2022; and the Persons-Rule era from October 7, 2022, the date the rule issued (its U.S.-persons licensing requirement took effect on October 12, the date marked in Figure 2). Each row estimates the within-board gap in the probability of an era-dated departure on that era's roster years (2018–20, 2020–22, and 2022–23, respectively). In Panel B the first row tests reversion of the 2023 within-board gap (one-sided against zero); the second and third rows compare the 2023 gap with the 2022 peak (two-sided) and with the mean of the 2020–21 gaps (one-sided; directional and not significant). The final two rows split the Persons-Rule era at China's border reopening of January 8, 2023, contrasting departures dated in the roughly three zero-COVID months (October 7, 2022 to January 7, 2023) with those dated in the reopened remainder of 2023, both on the 2022–23 roster years.

Table A10: Cumulative Departure of the 2019 Director Cohort

	(1) U.S.-nexus mean	(2) Boardmate mean	(3) Within-firm gap
Departed in 2020	0.157	0.137	$+0.0227^{**}$
			(0.041)
Departed in investment-sanctions era (2021–22)	0.328	0.284	$+0.0523^{***}$
			(<0.001)
Departed in Persons-Rule year (2023)	0.133	0.107	$+0.0369^{***}$
			(<0.001)
Departed by end-2023	0.618	0.528	$+0.1119^{***}$
			(<0.001)
Per-year escalation (Persons-Rule − investment era)	–	–	+0.0107
			(0.423)
Firm FE		Yes	
Director–firm pairs		50,693	
Firms (clusters)		2,645	
U.S.-nexus cohort members		1,052	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. The risk set is frozen at the 2019 rosters: every director serving in 2019 at a firm whose rosters are observed in each year 2020–2023 (50,693 director–firm pairs at 2,645 firms, of which 1,052 are U.S.-nexus). A director's departure year is the first year in which the director is absent from the firm's roster; re-appointments after an absence are ignored. Columns 1 and 2 report unconditional group means, and column 3 is the coefficient on U.S.-nexus in a linear-probability model with firm fixed effects, so it compares cohort members within the same firm. Because these probabilities condition only on 2019 membership, never on survival to a later year, the era rows are free of the survivor selection that attenuates late-era contrasts in the conditional hazard. The per-year escalation row compares the Persons-Rule year with the per-year rate of the two-year investment-sanctions era (the 2023 departure probability minus half the 2021–22 probability); it is positive and not statistically significant.

A.4.5 Pre-Trend Diagnostics

Our preferred mainland specification (2019–2024, Post $\geq$ 2021) has a deliberately thin pre-period. The single available pre-treatment coefficient (2020 relative to the 2019 base year) is -0.0003 (not significant at the 10% level), showing no detectable pre-shock divergence; parallelism cannot be assessed from one pre-period and is not the design's identifying assumption, with the within-firm placebo (Table 11) supplying the operative falsification against mean reversion. The event study shows a monotonic post-shock decline from -0.009 (2021) to -0.032^{***} (2024). The extended 2018–2024 mainland event study reveals a significant 2018 coefficient (-0.0063, significant at the 1% level), which Section 7 traces to the 124 early-treated firms; excluding them reverses the 2018 coefficient to a small significant positive. In Hong Kong, the surname-based event study shows significant pre-trend coefficients in 2015 (-0.030^{***}) and 2016 (-0.020^{**}), yet the relevant 2017–2019 pre-treatment window is clean (all not significant at the 10% level); the earlier anomaly likely reflects the onset of board internationalization among Hong Kong-listed firms in the mid-2010s. The nationality-based event study displays a similar pattern, with 2017–2019 clean (all not significant at the 10% level) and a monotonic post-shock decline reaching -0.029^{***} by 2024.

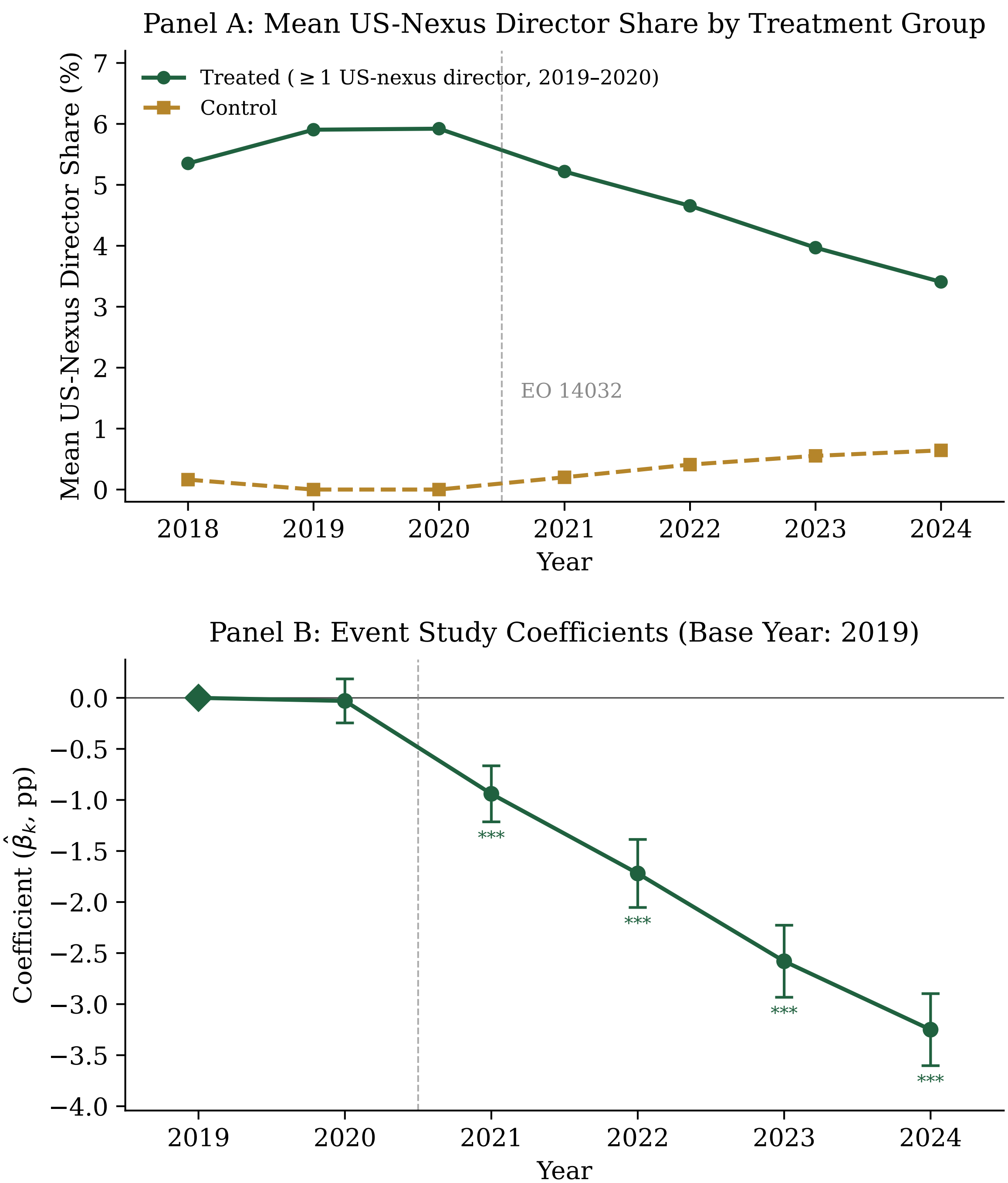

Figure A1: Event-study pre-trends, mainland A-share firms. Panel A plots the mean U.S.-nexus director share for treated ($\geq$ 1 U.S.-nexus director in 2019–2020) versus control firms; Panel B plots the year-by-treatment event-study coefficients on the preferred 2019–2024 window (base year 2019) with 95% confidence intervals, matching column 2 of Table 4. The single available pre-treatment coefficient (2020, not significant at the 10% level) shows no detectable divergence, and the monotonic post-2021 decline is consistent with a shock-driven exit; the design's causal weight rests on the within-firm and within-board comparisons set out in the text, which a one-year pre-window cannot supply.

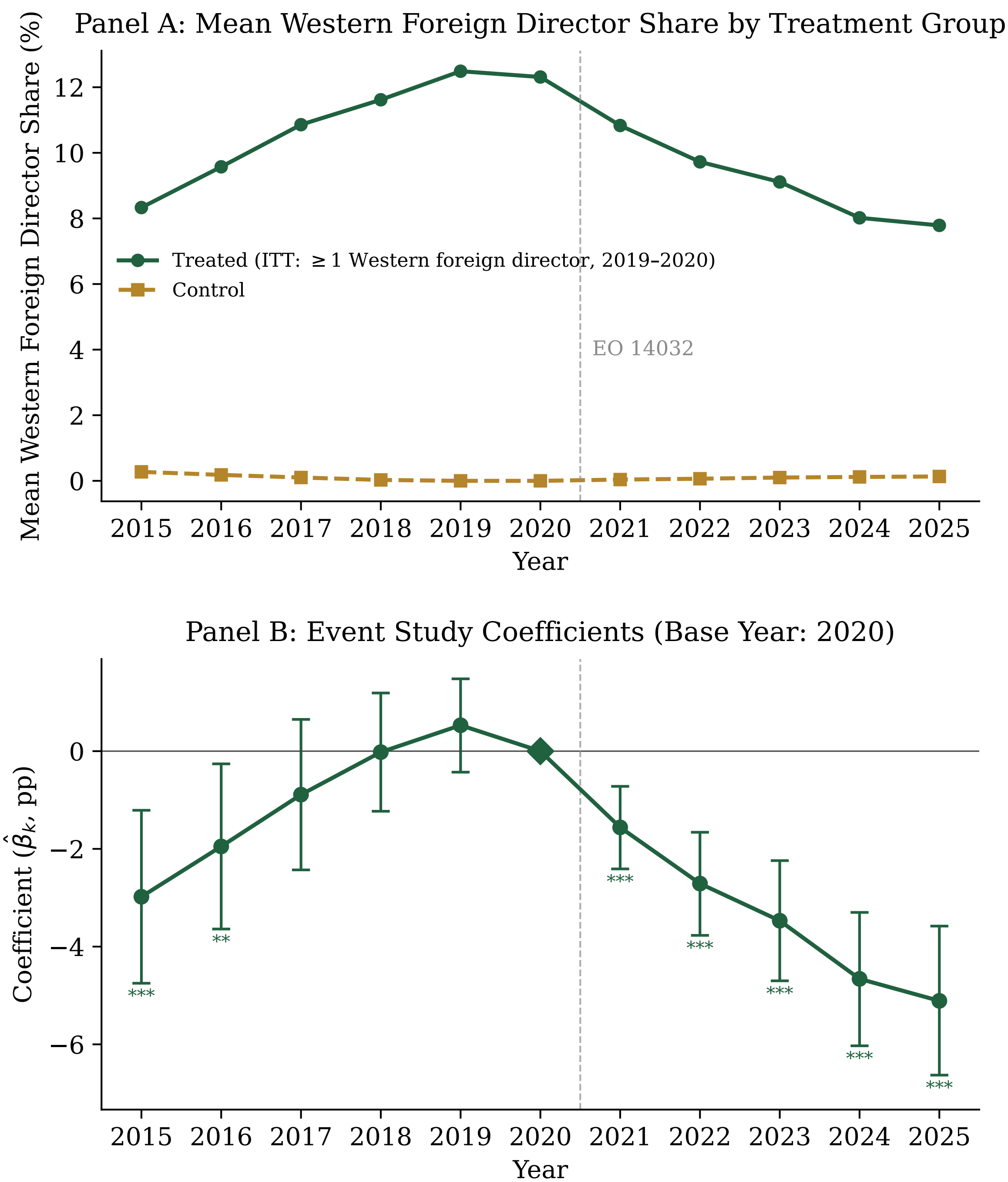

Figure A2: Event-study pre-trends, Hong Kong-listed Chinese firms. Panel A plots the mean Western-foreign director share for treated versus control firms; Panel B plots the year-by-treatment event-study coefficients (base year 2020) with 95% confidence intervals. The 2017–2019 pre-treatment window is clean (all not significant at the 10% level); the larger 2015–2016 coefficients reflect the onset of board internationalization in the mid-2010s rather than a sanctions response. The post-2021 decline mirrors the mainland pattern.

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